

EMBODIMENT ADAPTATION OF VISION-LANGUAGE-ACTION MODELS FROM PROPRIOCEPTIVE RESIDUALS

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ABSTRACT

Physical changes can make a pretrained vision–language–action (VLA) model’s commands produce the wrong robot motion even when the task remains feasible. We study embodiment adaptation from proprioceptive residuals: a calibrated command-to-motion predictor supplies a compact physical estimate that drives correction at the action interface while retaining the task backbone. Healthy calibration, controlled sensitivity probes, and a specified correction support precede deployment; subsequent updates need neither task-progress rewards nor a healthy reference trajectory. We compose residual calibration, partial correction support, and the actual sampled updates with a conditional execution-recovery bound. The metric covers the physical execution subsystem under the same supplied VLA commands; no VLA Jacobian or full-policy contraction certificate is required. Update holds, estimate age, clipping, realization error, and trajectory-dependent prediction error remain explicit. Necessary contact variables must be modeled or their effects bounded. Across four LIBERO simulation cohorts, correction improves paired success from 28/120 to 78/120, and from 34/120 to 68/120 under a preregistered held-out calibration (67% and 58% of matched healthy performance on the same scenarios); one robot calibration serves three VLA policies. Simulated ALOHA and GR1 use new calibration and prior identification; from a matched estimate an innovation-form update repairs ALOHA while adapting, so holding is needed only where the task margin is below the update’s own movement. Four preregistered checks refuted prior readings and are reported as primary. A constructed servo verifies the metric and envelope; on the evaluated robots the certificate is measured and splits, a humanoid arm certifying while the Panda under operational-space control has no same-command contraction, leaving its tube finite-horizon and its repair claim empirical.

1 INTRODUCTION

A VLA can select a useful action yet fail because the robot executes it differently from the conditions under which the policy was trained. Actuator offsets, reduced effectiveness, and increased friction alter the mapping from command to motion. Those changes also alter subsequent observations and policy decisions, creating a closed-loop physical distribution shift. We use *embodiment adaptation* to mean adaptation of this action-to-motion interface: structured physical OOD, rather than uncalibrated morphology transfer or missing visual and semantic competence.

Our mechanism is identification from proprioceptive prediction errors. A healthy command-to-motion model predicts the robot response; a calibrated sensitivity maps its residual into a small physical estimate; and that estimate changes how the next action is realized. LIBERO’s primary condition estimates six motion coordinates and corrects three rotation coordinates – counts of identified quantities and physical correction directions, not of editable neural weights.

Across four LIBERO suites, correction increases paired success from 28/120 to 78/120, and from 34/120 to 68/120 when the calibration is identified on held-out initial states; the long-horizon suite accounts for eight of the ten lost episodes. The same calibration serves three VLA policies. Torque bias and heavy friction recover under a corrected fault protocol; a locked joint is identified but not

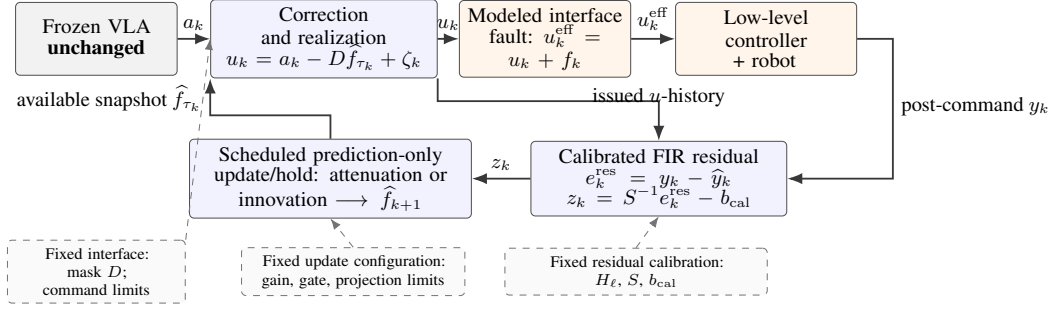


Figure 1: Evaluated prediction-only execution adapter. The frozen VLA emits a_k ; the available estimate snapshot is masked and realized before the modeled additive interface fault f_k . After execution, the calibrated FIR residual z_k drives the scheduled attenuation or innovation update. The task–observation loop is omitted; the theorem conditions on the supplied a_k . No nominal rollout, geodesic signal, task reward, or VLA update is used online (Appendix G.1).

repaired. ALOHA and GR1 use new calibration and prior fault identification, a different information budget from LIBERO’s within-episode adaptation.

From physical identification to recovery. Following selective neural fault recovery (Taheri et al., 2025), we connect residual calibration, adaptation, and physical execution. The nominal physical copy follows the same supplied commands, not a rerun healthy VLA; the execution state includes necessary robot, controller, object, and contact variables, and a reduced servo model must bound omitted interaction effects.

The evaluated chain is selected-coordinate estimation, applied mismatch, scheduled execution recovery, and predictor–trajectory feedback closure. Established same-input contraction supplies the propagation step; the contribution is its composition with calibration error, partial support, update holds, estimate age, and correction realization. Attenuation can converge to a biased target, innovation can reduce truth error without pairwise observer contraction, and the bounds preserve both mechanisms. Task success remains empirical or requires separate margin assumptions; the unevaluated composite extension is developed in the appendix.

2 PROPRIOCEPTIVE EMBODIMENT ADAPTATION

The evaluated policies are from the $\pi_0/\pi_{0.5}$, OpenVLA-OFT, and GR00T families (Black et al., 2024; 2025; Kim et al., 2025; Bjorck et al., 2025; NVIDIA, 2026).

2.1 EXECUTION INTERFACE AND CALIBRATION

At physical command step k , let a_k be the nominal action decoded by the retained VLA and D the fixed diagonal correction mask. Let $\hat{f}_k^{\text{app}} = \hat{f}_{\tau_k}$ be the estimate snapshot used for this command, with $0 \leq \tau_k \leq k$. The execution map includes the robot and its low-level controller. At the modeled interface, write

$$u_k = a_k - D\hat{f}_k^{\text{app}} + \zeta_k, \quad u_k^{\text{eff}} = u_k + f_k. \quad (1)$$

Here f_k is an additive physical offset in the interface’s command units. The discrepancy ζ_k includes imperfect realization of the intended correction and any clipping relative to that correction. Exact external subtraction with no clipping has $\zeta_k = 0$; editing a native decoder bias generally does not: a registered diagnostic on the native `action_out_proj/bias` edit measures a median realization error of 43% of the intended correction, because the decoder gain varies 27–41% across observations (Appendix E.6). Every published cohort subtracts the correction outside the network.

Fit a healthy finite-impulse-response (FIR) predictor, a sensitivity matrix S , and a healthy reference b_{cal} in inverse-sensitivity coordinates:

$$\hat{y}_k = \sum_{\ell=0}^K H_{\ell} u_{k-\ell} + \hat{c}, \quad e_k^{\text{res}} = y_k - \hat{y}_k, \quad z_k = S^{-1} e_k^{\text{res}} - b_{\text{cal}} = f_k + w_k. \quad (2)$$

The response y_k is acquired after executing u_k and updates the next estimate. The predictor uses the command actually presented at the modeled interface, before the unknown fault, with response timestamps aligned to that command. An internal intended correction is not interchangeable with the final decoder/clipping output. Healthy rollouts fit $H_{\ell}, \hat{c}$; controlled offset probes identify S . The error w_k contains history, model/sensitivity mismatch, noise, and imperfect healthy centering (Appendix G.2). The issued-command FIR buffer is observer memory; physical/controller delays remain in the execution state, while VLA internal memory belongs to the command source (Appendix B.4).

Table 1: Documented interface families. New robots receive new calibration. The listed update families follow the documented mathematical descriptions.

Interface	Predictor response	Offset estimate	Evaluation schedule
LIBERO Panda	Position increments; relative rotations ($K = 6$)	Observation attenuation	Within episode
ALOHA	Absolute joint positions ($K = 6$)	Observation attenuation	Prior identify, then hold
Fourier GR1	Absolute joint positions ($K = 6$)	Innovation normalization	Three prior episodes; held median

2.2 DOCUMENTED UPDATE AND CAUSAL TIMING

Let P_r select residual coordinates in the run’s declared units. Its normalization scale ρ is tied to those coordinates; changing units or whitening changes the estimator. Position increments and relative-rotation coordinates have different physical units; exact scaling must accompany each configuration. P_r is distinct from D : an uncorrected coordinate can still alter the shared gain; the recovered configurations use all six residual coordinates on LIBERO and all fourteen on ALOHA, while GR1 omits its unmodeled hand coordinates. LIBERO and ALOHA use observation attenuation, while GR1 uses innovation normalization:

$$\mathcal{T}_k^{\text{att}}(\hat{f}_k) = \Pi_{\mathcal{C}}[(1 - \gamma)\hat{f}_k + \gamma s_k z_k], \quad s_k = \frac{\mathbf{1}[\|P_r e_k^{\text{res}}\| > \delta]}{1 + \|P_r e_k^{\text{res}}\|^2 / \rho^2}, \quad (3)$$

$$\mathcal{T}_k^{\text{inn}}(\hat{f}_k) = \Pi_{\mathcal{C}}[\hat{f}_k + \alpha_k(z_k - \hat{f}_k)], \quad \alpha_k = \frac{\gamma}{1 + \|P_r S(z_k - \hat{f}_k)\|^2 / \rho^2}. \quad (4)$$

The set $\mathcal{C}$ bounds the estimate; the selected-coordinate results use a coordinatewise box. For a full-update flag $\chi_k \in \{0, 1\}$, apply

$$\hat{f}_{k+1} = \begin{cases} \mathcal{T}_k(\hat{f}_k), & \chi_k = 1, \\ \hat{f}_k, & \chi_k = 0. \end{cases} \quad (5)$$

A zero observation gate in Eq. (3) retains leakage when $\chi_k = 1$; $\chi_k = 0$ holds the entire estimate. The analysis treats both schedules explicitly. The snapshot τ_k separately captures delayed application to an action chunk, even if estimation is updated more often. LIBERO reference settings are $\gamma = .08$, $\rho = .15$, $\delta = .008$, and a $\pm .15$ estimate box in action coordinates.

Inputs: calibrated predictor and sensitivity; residual coordinates; mask D ; projection set; update family; gate and snapshot-application rule.

1. Obtain the next nominal action from the decoder or buffered chunk; select the available snapshot $\hat{f}_{\tau_k}$.
2. Realize the correction, apply limits, and execute. Save the actual command u_k at the modeled interface and its timestamp.
3. Associate the next proprioceptive response with that command and form e_k^{res} , z_k from the synchronized history and healthy reference.
4. Set χ_k using the declared schedule, update by Eq. (5), and store the estimate for later application.

Algorithm 1: Causal execution semantics used in the analysis. Fault truth is unavailable to the deployment estimator. The observed update/hold and application schedules must be logged to instantiate the schedule bounds.

2.3 CORRECTION SUPPORT, GAIN FAULTS, AND EVIDENCE SCOPE

The primary LIBERO condition estimates six coordinates and applies three rotation corrections. Translation/rotation masks were informed by fault-family diagnostics; the evaluations therefore use family-informed support and do not establish autonomous unknown-support selection. Three healthy episodes (285 steps) calibrate LIBERO; that calibration is reused across policies. ALOHA and GR1 use eight and six healthy episodes, respectively. Probe and fault-development data are additional to those healthy budgets.

For a matched diagonal FIR with constant actuator gain, define $\phi_{i,k} = \hat{y}_{i,k} - \hat{c}_i$ and $e_{i,k}^{\text{res}} = (g_i - 1)\phi_{i,k} + \varepsilon_{i,k}$. The gain branch uses gated recursive least squares and feasible inverse correction $a_i/\hat{g}_i$. Gains must remain bounded away from zero; joint gain/offset estimation additionally needs informative regressor variation.

A registered provenance pass closed the remaining implementation questions: every published cohort subtracts the correction outside the network, so native-edit realization error never enters the reported cells; the gate is a leakage deadzone on LIBERO and ALOHA and a full hold on GR1, and it never fired in the four headline samples or ALOHA’s identification episodes; $K = 6$ throughout. The realization and schedule terms remain in the analysis because the gated and native-edit regimes the map documents need them (Appendix G).

3 FROM RESIDUAL ESTIMATION TO EXECUTION RECOVERY

We analyze the evaluated prediction-only updates in the order in which errors reach the robot: selected-coordinate estimation, the applied correction, a scheduled execution tube, and predictor–execution coupling. The nominal comparison follows the *same supplied VLA commands*. Its contraction metric concerns the robot with its low-level controller; neither the VLA Jacobian nor a separately evolving healthy policy is required. The metric may be constructed from physical tracking dynamics (Appendix E.4) and, for the evaluated robots, was measured by same-command replay; that measurement is what decides whether the results below are certificates or finite-horizon statements (Appendix G.15). Prediction-only adaptation does not compute the nominal response or a geodesic online.

The execution state must close the physical transition: it contains robot and necessary low-level controller, delay, and interpolation states, plus object/contact states when their evolution affects execution. Omitted contact or environment effects require a uniform disturbance bound on the stated domain; otherwise the certificate applies only to the reduced servo model (Appendix B.4). VLA internal memory and the separate observer FIR history are excluded unless a retained physical dependence requires additional state or bounded forcing (Appendix B.4).

Same-input contraction and disturbance-to-tracking bounds are established machinery (Lohmiller & Slotine, 1998; Manchester & Slotine, 2017). Our specialization composes residual calibration with partial correction support, update holds, snapshot age, clipping and realization error, and a prediction-dependent small-gain condition. The optional continuous composite controller and fixed-data gradient candidate are developed only in Appendices D and B.3.

3.1 SELECTED-COORDINATE ESTIMATION

Physical command steps are indexed by k ; a_k denotes the supplied VLA command and e_k^{res} the measured proprioceptive residual.

The calibrated FIR regression gives

$$z_k = f_k + w_k, \quad w_k = S^{-1} \left[\sum_{\ell=0}^K G_\ell (f_{k-\ell} - f_k) + (G_\Sigma - S)f_k + \Delta b + n_k \right]. \quad (6)$$

$G_\Sigma = \sum_\ell G_\ell$; Δb is calibration bias and n_k is predictor mismatch. The history term persists for K steps after a constant fault jumps. Its bound must hold on corrected trajectories.

Proposition 1 (Selected-coordinate target error). *Let $\mathcal{C}$ be a coordinate box, D a fixed binary diagonal mask, $f_k, \hat{f}_0 \in \mathcal{C}$, $E_k^D = \|D(\hat{f}_k - f_k)\|$, $\epsilon_k^D \geq \|Dw_k\|$, and $\nu_k^D \geq \|D(f_{k+1} - f_k)\|$. For the actual full-update flag $\chi_k \in \{0, 1\}$, $0 < \gamma \leq 1$, and $s_k, \alpha_k \in [0, 1]$, attenuation and innovation respectively satisfy*

$$E_{k+1}^D \leq (1 - \chi_k \gamma) E_k^D + \chi_k \gamma [(1 - s_k) \|Df_k\| + s_k \epsilon_k^D] + \nu_k^D, \quad (7)$$

$$E_{k+1}^D \leq (1 - \chi_k \alpha_k) E_k^D + \chi_k \alpha_k \epsilon_k^D + \nu_k^D. \quad (8)$$

Box projection is nonexpansive on the selected coordinates; a coupled convex set need not have this property. With $D = I$, both bounds hold for any closed convex set. The normalizer selection P_r remains distinct from D : uncorrected coordinates can still change s_k or α_k . A full hold has $\chi_k = 0$; target-only gating has $\chi_k = 1$, $s_k = 0$ and retains leakage.

Target convergence is not pairwise contraction. For fixed noiseless data and attenuation s , the legacy map contracts toward $\Pi_{\mathcal{C}}(sf)$, generally a biased target. For the scalar interior innovation map $T(h) = h + \gamma(z - h)/[1 + (z - h)^2/\rho^2]$, with $v = (z - h)^2/\rho^2$,

$$T'(h) = 1 + \gamma \frac{v - 1}{(1 + v)^2}, \quad \max_{v \geq 0} T'(h) = 1 + \gamma/8. \quad (9)$$

At $\gamma = .08$, $v = 3$, the truth-error multiplier is .98 but the local pairwise multiplier is 1.01. Thus Eq. (8) does not establish global Euclidean observer contraction.

3.2 APPLIED CORRECTION AND PHYSICAL MISMATCH

Propagate $B_k^D \geq E_k^D$ using the applicable recursion. For the snapshot $\hat{f}_{\tau_k}$ used at step k , $0 \leq \tau_k \leq k$,

$$q_k = (I - D)f_k + D(f_k - \hat{f}_{\tau_k}) + \zeta_k, \quad (10)$$

$$B_k^{D, \text{app}} = \min \left\{ B_k^D + \sum_{j=\tau_k}^{k-1} \|D(\hat{f}_{j+1} - \hat{f}_j)\|, B_{\tau_k}^D + \sum_{j=\tau_k}^{k-1} \nu_j^D \right\}. \quad (11)$$

Here $\|D(\hat{f}_{\tau_k} - f_k)\| \leq B_k^{D, \text{app}}$ and ζ_k includes native-decoder realization and command clipping. Projection of an estimate does not ensure feasible commands.

3.3 SCHEDULED EXECUTION RECOVERY

The following result is stated directly in execution distances, so the certificate need not be derived from a differentiable transition. A smooth, command-independent Riemannian metric is one sufficient construction (Appendix B.4). Contact and history effects must be included in the execution state or in the stated transition remainder.

Theorem 2 (Scheduled same-command execution tube). *Fix the supplied command sequence a_k . Let $\xi_{k+1}^{\text{nom}} = F_{\text{exec}, k}^0(\xi_k^{\text{nom}}, a_k)$ be its nominal physical response, and let ξ_k be the realized closed or reduced execution state. Let d_k be metrics on declared execution domains Ω_k , chosen independently of the supplied command and adapted estimate. Assume a declared, time-independent physical baseline metric δ_{phys} and constants $0 < c_- \leq c_+ < \infty$ satisfy, uniformly over k and the certified domains,*

$$c_- \delta_{\text{phys}}(\xi, \xi') \leq d_k(\xi, \xi') \leq c_+ \delta_{\text{phys}}(\xi, \xi'). \quad (12)$$

For $\lambda_k^{\text{exec}}, L_k, \eta_k \geq 0$, suppose, uniformly over allowed commands a and admissible $\xi, \xi' \in \Omega_k$,

$$d_{k+1}(F_{\text{exec},k}^0(\xi, a), F_{\text{exec},k}^0(\xi', a)) \leq \lambda_k^{\text{exec}} d_k(\xi, \xi'), \quad (13)$$

and, uniformly over admissible realized and omitted states,

$$d_{k+1}(\xi_{k+1}, F_{\text{exec},k}^0(\xi_k, a_k)) \leq L_k \|q_k\| + \eta_k. \quad (14)$$

Then, for every admissible $\xi' \in \Omega_k$,

$$d_{k+1}(\xi_{k+1}, F_{\text{exec},k}^0(\xi', a_k)) \leq \lambda_k^{\text{exec}} d_k(\xi_k, \xi') + L_k \|q_k\| + \eta_k, \quad (15)$$

$$R_{k+1} = \lambda_k^{\text{exec}} R_k + L_k (\|(I - D)f_k\| + B_k^{D,\text{app}} + \bar{\zeta}_k) + \eta_k \quad (16)$$

bounds $X_k := d_k(\xi_k, \xi_k^{\text{nom}})$ from $R_0 \geq X_0$ and gives $\delta_{\text{phys}}(\xi_k, \xi_k^{\text{nom}}) \leq R_k/c$, provided $\|\zeta_k\| \leq \bar{\zeta}_k$, the supplied commands are admissible, and all states and tubes needed to invoke Eqs. (13)–(14) remain in the declared domains.

Equation (13) is the certificate itself and may be verified directly for a nonsmooth sampled or hybrid transition. The Jacobian metric inequality in Eq. (45) is a smooth-mode sufficient condition, not an additional theorem assumption. This is recovery of physical execution under the same supplied commands, not an independently rerun healthy VLA trajectory. No VLA Jacobian or policy-state metric is needed. Arbitrary finite $\lambda_k^{\text{exec}} \geq 0$ give a finite-horizon bound; exponential forgetting requires $\prod_{i=j}^{k-1} \lambda_i^{\text{exec}} \leq C\varrho^{k-j}$ on every subinterval, $C \geq 1$, $\varrho < 1$. Hybrid events and intermediate chunk motion must be included in the direct transition conditions. Task preservation is a separate conditional extension (Appendix B.4). The contraction and transition constants were subsequently measured on the evaluated robots rather than left open: Eq. (13) holds on the GR1 arm servos and fails on the Panda under operational-space control, where the measured ratio is 1.00 and this theorem therefore yields only its finite-horizon form (Appendix G.15). Task margins remain uncertified.

3.4 CLOSING PREDICTION–EXECUTION COUPLING

The predictor is evaluated on trajectories changed by the correction. A trajectory-dependent residual bound therefore requires a coupled condition.

Corollary 3 (Active-every-step recovery region). *With $\chi_k = 1$, $\tau_k = k$, suppose*

$$X_{k+1} \leq \lambda_d X_k + bE_k^D + d_0, \quad \|Dw_k\| \leq \epsilon_0 + k_E E_k^D + k_X X_k. \quad (17)$$

For nonnegative coefficients, let $a = 1 - \lambda_d > 0$, $c = 1 - k_E > 0$, $\Delta = ac - bk_X > 0$ and masked drift at most ν . Set $v = \epsilon_0 + \bar{a}_{\text{att}} + \nu/\gamma$ for attenuation, where $\bar{a}_{\text{att}} \geq (1 - s_k)\|Df_k\|$; set $v = \epsilon_0 + \nu/\underline{\alpha}$ for innovation with $\alpha_k \geq \underline{\alpha} > 0$. Any rectangle containing (X_0, E_0^D) , inside the domain, and satisfying

$$a\bar{X} \geq b\bar{E} + d_0, \quad c\bar{E} \geq k_X \bar{X} + v \quad (18)$$

is forward invariant.

Here $b = L$ for the masked interface bound; predictor-error growth k_X closes the feedback between correction and visited states. The condition $ac > bk_X$ requires decay of the two subsystems to dominate their mutual error growth. Appendix B.6 gives the equality corner and a sufficient startup budget requiring valid error and execution-domain bounds.

3.5 AN ANALYTICALLY VERIFIED EXECUTION INSTANCE

For the servo in Appendix F, a constant metric $P = \text{diag}(1, 4)$ satisfies $A^\top P A = .92^2 P$ for the nominal robot–controller map, giving $\lambda_k^{\text{exec}} = .92$ and $L_k = 2$. It therefore verifies the smooth sufficient condition and the direct contraction bound used by Theorem 2. Figure 2 propagates the same estimator and execution bounds through fault onset, full-update holds, and delayed snapshots. The limiting metric errors are 14.94 mm for attenuation and 5.64 mm for innovation, inside bounds of 33.29 and 9.01 mm, respectively. The construction has a matched linear predictor and fixed centering bias; it verifies the scheduled execution tube, with no trajectory-dependent predictor mismatch or contact transition to certify. Manipulation requires the physical closure and domain conditions stated above.

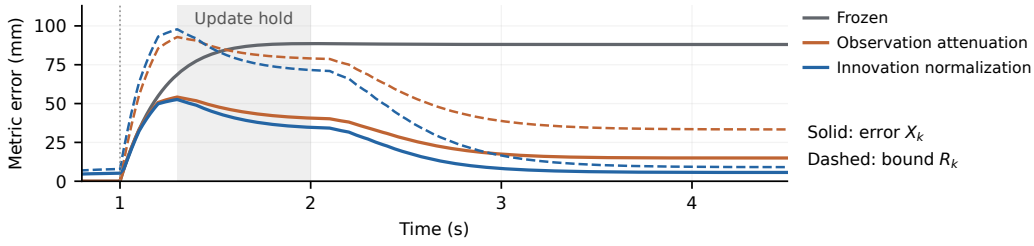


Figure 2: Recovery in the analytically verified execution servo. Solid curves show metric tracking error X_k ; dashed curves show its propagated bound R_k . The fault starts at 1 s; shading marks a full-update hold. This constructed example uses the documented sampled update families and fixed centering bias, not measured VLA trajectories (Appendix F).

4 EXPERIMENTAL EVIDENCE

Protocol and scope. All task-success results are in simulation. LIBERO (Liu et al., 2023) adapts within episodes; ALOHA (Zhao et al., 2023) and GR1 identify on prior episodes. Comparisons pair task and initial state; policy sampling is not seed-pinned, and repeated $n=20$ cells vary by up to ± 2 successes (Appendix G.3). *Fixed/broken* count failure-to-success/success-to-failure changes. Implementation provenance was recovered for all nine cohorts, and the rotation mask is reproduced by a preregistered healthy-only gate (Appendix G.1). Every comparison below was preregistered with its refutation condition written first, and the refutations are reported as primary.

4.1 EPISODE-LOCAL RECOVERY ACROSS FOUR SUITES

Table 2: Primary comparison: $\pi_{0.5}$ on LIBERO, uniform $+0.05$ six-axis offset; six coordinates estimated, three rotation corrections applied. Blocks: calibration probed at the evaluated initial states (shipped) and preregistered held-out calibration from disjoint states. Matched healthy controls on the same 120 scenarios: 116–118/120 frozen; the law on the healthy robot changes $+1/0$ net episodes, $p = 1.0$.

Suite	Shipped calibration				Held-out calibration			
	Frozen	Corr.	f/b	Exact p	Frozen	Corr.	f/b	Exact p
Spatial	8/20	18/20	10/0	.00195	9/20	18/20	9/0	.00391
Goal	15/40	29/40	15/1	.000519	18/40	28/40	11/1	.00635
Object	5/20	16/20	11/0	.000977	7/20	15/20	10/2	.0386
Long (10)	0/40	15/40	15/0	$6.1 \cdot 10^{-5}$	0/40	7/40	7/0	.0156
Pooled	28/120	78/120	51/1	$2.4 \cdot 10^{-14}$	34/120	68/120	37/3	$2.0 \cdot 10^{-8}$

With the shipped calibration, success increases from 23.3% to 65.0% (paired gain 41.7 points; approximate 95% interval $[32.5, 50.8]$ under an episode-iid model), all four suites significant after Bonferroni correction. The held-out replication was registered with ± 3 reproduction bands: three suites reproduce within band; the long-horizon suite is *refuted*, dropping to 7/40 against a registered threshold of five. A task-clustered bootstrap leaves the pooled effect intact under both calibrations ($[+36, +65]$ and $[+20, +49]$ episodes), and three of four suites individually significant at cluster level under the shipped calibration against one of four under the held-out one. The honest headline is therefore two numbers – 78/120 and 68/120, or 67% and 58% of matched health – because the measured sensitivity varies with the probed initial state by up to a factor of two in single axes (Appendix G.4) and long-horizon episodes give that error 200+ steps to accumulate.

4.2 WHERE CORRECTION WORKS, WHERE IT FAILS, AND THE INTERFACE CEILING

Translation offset 0.15 improves 4/20 $\rightarrow$ 19/20, while uniform offset 0.15 reaches only 2/20 with translation left uncorrected; mixed-sign rotation improves 2/12 $\rightarrow$ 10/12. Under changing offsets

correction still improves task success despite waveform RMS errors of 45–79% of fault amplitude, and gain faults $g = .30$ and $.20$ both reach 17/20, from 1/20 and 0/20 (Appendices G.5–G.8, Figure 5).

The measured residual exposes a limitation beyond matrix conditioning: for true x -offset $+0.05$ the separated x estimate is -0.0130 under a uniform six-axis fault but $+0.0244$ under a single-axis one, so an approximately condition-3 healthy sensitivity does not ensure accurate reconstruction on faulted trajectories (Appendix G.2).

Table 3: Disturbances below the Panda operational-space controller; friction, lock, and healthy rows use the corrected reset protocol, the torque row the historical one (Appendix G.10). Translation correction with healthy-bias subtraction.

Physical intervention	Frozen	Corrected	Fixed	Broken
Elbow torque bias 5 N m	20/40	32/40	15	3
High elbow friction (+20), $n=40$	0/40	13/40	13	0
Elbow lock within ± 0.05 rad	0/20	0/20	0	0
Healthy elbow, law running	19/20	20/20	1	0

The friction rerun was registered before running ($p = 2.4 \cdot 10^{-4}$), and the lock cell reproduces exactly, estimate included. Its residual sits 15–23 healthy standard deviations from baseline with no recovered success: detection does not establish correction authority – a limit of this interface, not an impossibility result for replanning or other controllers.

Registered baselines and the ceiling. On the headline fault with full logs, the method reproduces to the episode (6 $\rightarrow$ 18 of 20 spatial, 0 $\rightarrow$ 15 of 40 long-horizon; frozen rates move with unpinned sampling). A calibrated *static* observer ($K = 0$) matches it on the short suite; its six-of-forty long-horizon gap ($p = .07$) did not survive a seed-pinned rerun at $n = 80$ (10/80 versus 9/80, $p = 1.0$), so FIR memory is *refuted* as a resolvable advantage on that suite. The *innovation* law lands within three episodes of the deployed law on both cohorts and removes the attenuation bias on the well-identified rotation channels. *Known-fault correction through the same interface* reaches 19/20 and 22/40: the estimated correction attains 1.09 and 0.71 of its net gain. The long-horizon shortfall localizes to r_y , and its cause is measured: plants fitted to the closed-loop healthy log recover an r_y DC gain near .10 against the probe’s .28, and on all three rotation channels the settled estimate tracks that ratio (Appendix G.15). With all six channels corrected exactly, spatial reaches 20/20 against a healthy 19/20, so the fault is fully cancellable through this interface and the rotation-only mask costs at most one episode of ceiling. Against the best raw-error integral gain, run paired and seed-pinned on one scenario set, the method wins 18/20 to 3/20 (15 fixed, 0 broken, $p = 6.1 \cdot 10^{-5}$): the integral’s x estimate never arrives and its z overshoots (Appendices G.8–G.9).

4.3 POLICY REUSE, NEW ROBOTS, AND WHAT HOLDING BUYS

The LIBERO predictor and sensitivity fitted with $\pi_{0.5}$ are reused unchanged with OpenVLA-OFT and GR00T N1.7: on spatial rotation, translation and gain faults, OFT reaches 11, 14 and 17 of 20 from zero, and N1.7 14, 15 and 19 from 0, 2 and 0 – calibration reuse between policies on one robot. ALOHA and GR1 receive robot-specific calibration.

Table 4: Reported *prior identification plus hold* evaluations. Extra fault-identification episodes are additional to healthy calibration and evaluation. Offsets are .02 rad on ALOHA and .10 rad on GR1.

Robot / policy	Task	Healthy	Frozen	Held	Extra identification
ALOHA / π_0	Cube	17/40	0/40	15/40	1 episode
GR1 / N1.5	Plate	22/30	1/30	19/30	3 episodes
	Tray	17/30	1/30	13/30	3 episodes

A registered comparison then equalized the information budget: every arm starts from the *same* stored estimate and differs only after step 0. On ALOHA the innovation law updating throughout repairs

14/40 against the held 15/40 ($p = 1.0$), while the deployed attenuation law repairs 0/40, its estimate pulled to .016 rad within twenty steps by the attenuated fixed point and deadzone leakage – a 0.42 cm standing error at the grasp. Failure under continued adaptation is that law’s bias, not updating as such. On GR1 the innovation law loses five of thirty to updating (22/30 against 17/30), so holding is required where the task margin is smaller than the update’s own movement (Appendix G.11). Healthy adaptation changes +1/0 net of 120 ($p = 1.0$).

The execution certificate, measured. Figure 2 instantiates the same-command bound on a constructed servo; on the evaluated robots the certificate was then measured, and splits. The GR1 right arm certifies as a first-order servo (poles .72–.77); ALOHA at 50 Hz does not. The Panda under OSC_POSE certifies per-axis tracking yet has *no same-command contraction*: replaying identical commands from perturbed states gives a per-step gap ratio of 1.00 in all four preregistered metrics, because the controller sets each goal relative to the current pose. Theorem 2 is therefore finite-horizon only there – propagated with the measured constants it covers every step but ends near $100\times$ the largest measured error – and the small-gain condition fails through $a = 1 - \lambda$, not through the coupling constants, which are small (Appendix G.15).

Evidence boundary. These results support correction of the tested execution shifts; task margins, arbitrary perceptual OOD, and deadlines remain uncertified. Adapter compute measures .17/.46/.77 ms (median/p99/max) per control step against a policy call near 900 ms (Appendix G.14). Sustained recovery time is not reported against a decay rate, the measured $\lambda \approx 1$ implying none here. X2 hardware results are sensing diagnostics; paired physical-robot repair is unevaluated.

5 RELATED WORK

VLA adaptation and residual recovery. J-PARC is the closest architectural comparison: a frozen VLA, joint-motion history, inferred fault context, and learned residual action calibration (Jo et al., 2026), trained with fault labels and paired healthy/faulted information. Our distinction is online numerical identification from calibrated proprioceptive residuals, without training a fault-conditioned residual policy; healthy calibration, sensitivity probes, and fault-family development remain part of our data budget. TT-VLA uses test-time reinforcement learning with a pretrained progress estimator (Liu et al., 2026), BORA learns a residual through reinforcement learning (Chen et al., 2026), and VLA-Corrector uses visual prediction during inference (Pan et al., 2026); Residual Control for Fast Recovery from Dynamics Shifts keeps a nominal policy fixed and learns a bounded, tracking-driven residual (Jayasinghe et al., 2026). Our update requires neither task-progress reward nor tracking feedback, and a matched J-PARC comparison is needed to establish an empirical advantage.

Adaptive and contraction-based control. Disturbance observers and robust adaptation underpin estimation (Chen et al., 2016; Ioannou & Sun, 1996); composite control combines tracking and prediction (Slotine & Li, 1989), and informative stored data support parameter convergence (Chowdhary & Johnson, 2010; Parikh et al., 2019). Neural-Fly combines learned representations with compact composite adaptation (O’Connell et al., 2022), RMA learns legged-dynamics adaptation (Kumar et al., 2021), and fault recovery depends on surviving authority (Tao et al., 2001). Contraction gives differential trajectory and perturbation bounds (Lohmiller & Slotine, 1998; Tsukamoto et al., 2021b); geodesic adaptive feedback and neural metric constructions are established (Lopez & Slotine, 2020; Tsukamoto et al., 2021a), and parameter-dependent metrics need extra transient terms (Gessow & Lopez, 2023). We specialize this foundation to physical execution with prediction-only adaptation, biased targets, limited support, native realization, and delayed application; the composite extension is unevaluated in the reported VLA trials.

6 DISCUSSION AND LIMITATIONS

The execution metric concerns the robot and its low-level controller under shared commands, and now has a measured answer: available on a humanoid arm, absent on the Panda under operational-space control. Manipulation further requires closed object/contact dynamics or bounds on omitted contact effects; useful commands, task success, and unmodeled perceptual or contact-topology change stay

outside the guarantee. Measuring the theory's own constants separated its two halves: the estimator recursions are predictive on the deployed plant, while its execution certificate is unavailable there.

REPRODUCIBILITY STATEMENT

Sources, proofs, count ledgers, figure scripts, constructed checks, logging schemas, calibration arrays, and per-run configuration records are included. The primary LIBERO cells were independently rerun under a corrected reset protocol and reproduce to the episode; historical traces for the remaining legacy cohorts are absent, and the implementation manifest lists what their records recover.

AI USE STATEMENT

Claude assisted earlier coding and writing; ChatGPT assisted revision, mathematical analysis, literature checks, and constructed examples. The authors are responsible for the proofs, code, and empirical claims.

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A IDENTIFIABILITY AND SCOPE OF THE SAMPLED GUARANTEES

The main text states the update/hold recursions, delayed-application bound, finite-horizon tube, and its active-every-step invariant-region specialization. The following restriction explains the independent bias calibration they need.

Proposition 4 (Additive ambiguity). *In $y_k = S(u_k + f) + b_y + \varepsilon_k^{\text{obs}}$, with constant f, b_y and invertible S , every pair $(f + v, b_y - Sv)$ produces the same observations. Without an independent restriction on b_y , only $Sf + b_y$ is identifiable.*

Healthy calibration or a known healthy prefix supplies a bias reference, subject to history and modeling error. With known nonsingular sensitivity, additive fault observation does not require new command excitation. Gain identification has a different regressor and information requirement. For convenience the sampled analysis uses

$$z_k = f_k + w_k, \quad \|w_k\| \leq \epsilon_k, \quad E_k = \|\hat{f}_k - f_k\|, \quad \|f_{k+1} - f_k\| \leq \nu_k. \quad (19)$$

Bounds on w_k must hold on corrected trajectories. A mean healthy residual does not itself establish a bound in new contact modes. Likewise, the local incremental inequality in Eq. (15) concerns the physical execution map under the same supplied VLA commands (Appendix B.4). The certificate includes necessary low-level controller state, but does not require VLA internal memory or a VLA Jacobian. A healthy task rollout generally supplies different commands. The proofs and quantitative error budget follow in Appendix B.

B CONDITIONAL ESTIMATION AND RECOVERY GUARANTEES

The results below describe the conditions under which a calibrated residual can support execution recovery. The metric concerns robot dynamics and their low-level controller under a shared supplied command stream. It is not a metric for the VLA feedback loop, and the reported robot experiments do not verify this physical metric. In particular, the estimator results and the trajectory results have different assumptions. All norms are Euclidean or their induced operator norms unless stated otherwise.

B.1 RESIDUAL HISTORY, CALIBRATION ERROR, AND IDENTIFIABILITY

Let u_k be the actual command sent by the adapter before the additive fault. An idealized local FIR description is

$$y_k = \sum_{\ell=0}^K G_\ell(u_{k-\ell} + f_{k-\ell}) + c + b_y + \varepsilon_k^{\text{FIR}}, \quad \hat{y}_k = \sum_{\ell=0}^K H_\ell u_{k-\ell} + \hat{c}. \quad (20)$$

Here b_y is a constant healthy residual bias in measurement units, and $\varepsilon_k^{\text{FIR}}$ includes departures from the local FIR description. Define $G_\Sigma = \sum_{\ell} G_\ell$, a calibrated nonsingular sensitivity S , a healthy bias calibration $\hat{b}_y = Sb_{\text{cal}}$, and

$$z_k = S^{-1}(y_k - \hat{y}_k - \hat{b}_y) = f_k + w_k, \quad (21)$$

$$w_k = S^{-1} \left[\sum_{\ell=0}^K G_\ell(f_{k-\ell} - f_k) + (G_\Sigma - S)f_k + (b_y - \hat{b}_y) + n_k \right], \quad (22)$$

$$n_k = \varepsilon_k^{\text{FIR}} + \sum_{\ell=0}^K (G_\ell - H_\ell)u_{k-\ell} + (c - \hat{c}).$$

Thus $e_k^{\text{res}} - \hat{b}_y \simeq Sf_k$ is a steady or slowly varying approximation, not an instantaneous identity at fault onset. If $\|n_k\| \leq \bar{n}$, $\|b_y - \hat{b}_y\| \leq \Delta_b$, $\|G_\Sigma - S\| \leq \Delta_S$, $\|f_k\| \leq F$, and $\|f_{k+1} - f_k\| \leq \nu$, then

$$\|w_k\| \leq \epsilon := \|S^{-1}\| \left(\bar{n} + \Delta_b + \Delta_S F + \nu \sum_{\ell=0}^K \ell \|G_\ell\| \right). \quad (23)$$

This follows by $\|f_{k-\ell} - f_k\| \leq \ell\nu$ and the triangle inequality. At an abrupt change, the exact history term in (22) is preferable to this uniform bound; for a constant post-onset fault it vanishes after the FIR window has filled. An incorrectly initialized history contributes to n_k until that history has been replaced. Bounds on predictor error and sensitivity mismatch must hold on the corrected trajectories, not only on healthy calibration data.

Proof of Proposition 4 and its scope. In the constant-sensitivity model $e_k^{\text{res}} = Sf + b_y + \varepsilon_k^{\text{obs}}$, with constant unknown f and unrestricted constant unknown b_y , f and b_y cannot be separately identified. For every vector v , the pair $(f + v, b_y - Sv)$ gives exactly the same observations. With known b_y and nonsingular S , the noiseless residual does determine f . The identity $S(f + v) + (b_y - Sv) = Sf + b_y$ proves the ambiguity. For known bias, $f = S^{-1}(e_k^{\text{res}} - b_y)$ in the noiseless model.

A known healthy pre-onset interval can therefore provide the bias calibration within the same episode, subject to noise and history transients. More generally, a known fault subspace or time-varying sensitivity can change identifiability; the proposition is not an impossibility statement for every sensor and model. An offline mean of the raw healthy residual estimates b_y only when the remaining healthy modeling error is zero-mean under that sampling distribution. Otherwise it estimates an aggregate healthy offset whose transfer to corrected trajectories needs its own error bound. A mean of the settled, attenuated legacy estimate is a different quantity and must not be silently substituted for this raw residual calibration. For example, if a constant healthy observation in fault units is b_0 and attenuation is fixed at s_0 , a settled legacy estimate equals $s_0 b_0$ when projection is inactive. Subtracting that estimate before attenuation leaves $s_0(b_0 - s_0 b_0) = s_0(1 - s_0)b_0$, generally nonzero. The averaging quantity, units, and subtraction order therefore affect the identification error and must match the executed implementation.

B.2 THE TWO ADDITIVE UPDATE LAWS AND THE PHYSICAL COMMAND CLOCK

Let $C \subset \mathbb{R}^d$ be closed and convex, assume $f_k \in C$, and let Π_C denote Euclidean projection. Let $\hat{f}_0 \in C$ and $E_k = \|f_k - \hat{f}_k\|$. The box constraint in the experiments is one choice of C ; it bounds the estimate but does not by itself guarantee safe plant motion.

Full-error counterpart of Proposition 1. Suppose (21) holds with $\|w_k\| \leq \epsilon_k$ and $\|f_{k+1} - f_k\| \leq \nu_k$. First consider one active innovation update

$$\hat{f}_{k+1} = \Pi_C\{\hat{f}_k + \alpha_k(z_k - \hat{f}_k)\}, \quad 0 \leq \alpha_k \leq 1. \quad (24)$$

Then

$$E_{k+1} \leq (1 - \alpha_k)E_k + \alpha_k\epsilon_k + \nu_k. \quad (25)$$

Writing $P_{j,k} = \prod_{i=j}^{k-1} (1 - \alpha_i)$ with empty products equal to one gives the finite-horizon bound

$$E_k \leq P_{0,k}E_0 + \sum_{j=0}^{k-1} P_{j+1,k}(\alpha_j\epsilon_j + \nu_j). \quad (26)$$

If $\epsilon_k \leq \epsilon$, $\nu_k \leq \nu$, and $\alpha_k \geq \underline{\alpha} > 0$, then, with $q = 1 - \underline{\alpha}$,

$$E_k \leq q^k E_0 + \epsilon + \frac{\nu}{\underline{\alpha}}. \quad (27)$$

In particular, an exactly modeled constant fault converges exponentially.

Proof. Since $f_k \in C$, projection nonexpansiveness gives

$$\begin{aligned} \|\hat{f}_{k+1} - f_k\| &\leq \|(1 - \alpha_k)(\hat{f}_k - f_k) + \alpha_k w_k\| \\ &\leq (1 - \alpha_k)E_k + \alpha_k\epsilon_k. \end{aligned}$$

Adding $\|f_{k+1} - f_k\|$ proves the recursion; induction proves the product bound. Also $P_{0,k} \leq q^k$, $\sum_{j=0}^{k-1} \alpha_j P_{j+1,k} = 1 - P_{0,k} \leq 1$, and $\sum_{j=0}^{k-1} P_{j+1,k} \leq \sum_{i=0}^{k-1} q^i \leq 1/\underline{\alpha}$, proving the uniform bound. $\square$

The innovation normalizer corresponds to

$$\alpha_k = \frac{\gamma}{1 + \|P_r S(z_k - \hat{f}_k)\|^2 / \rho^2}, \quad 0 < \gamma \leq 1. \quad (28)$$

If C has diameter D_C and $\|w_k\| \leq \epsilon$, then $\underline{\alpha} = \gamma / [1 + \|P_r S\|^2 (D_C + \epsilon)^2 / \rho^2]$ is a valid, possibly conservative lower bound. This estimator does not require online persistent excitation when the full-rank, constant S is already known: the fault is directly observed through S . Estimating S from healthy data, and estimating multiplicative faults, are separate identification tasks.

Deadzone placement. Equation (26) remains valid when a gate sets $\alpha_k = 0$. The uniform exponential bound then requires a positive lower bound on accumulated effective gain over every fixed-length sliding window, not merely a positive ungated gain or some active updates. For a full-channel innovation gate that skips updates when $\|S(z_k - \hat{f}_k)\| \leq \delta$, an inactive step implies

$$E_k \leq \epsilon + \|S^{-1}\| \delta. \quad (29)$$

This full-error implication requires the full invertible S . A selected gate $\|P_r S(z_k - \hat{f}_k)\| \leq \delta$ gives it only when $P_r S$ is injective on the estimated fault space, with $\|S^{-1}\|$ replaced by $1/\sigma_{\min}(P_r S)$. A rank-deficient selection cannot bound the unobserved error components. For constant f and bounded noise, this ball is forward invariant under active steps, and distance above it decreases by at least the ungated factor q on every active step. Indeed, the inactive-step bound follows directly from $z_k - \hat{f}_k = f_k - \hat{f}_k + w_k$; for active steps, (25) maps that ball into itself and contracts any excess. A raw-residual gate has no corresponding error implication: after a fault disappears, e_k^{res} can be small while $\hat{f}_k$ is not. The distinct implementation that zeroes the observation but retains the $-\hat{f}_k$ leakage decays on inactive steps rather than freezing. These gate conventions must be reported separately.

Proposition 5 (Attenuation in the legacy update). *For*

$$\hat{f}_{k+1} = \Pi_C\{(1 - \gamma)\hat{f}_k + \gamma s_k z_k\}, \quad 0 < \gamma \leq 1, \quad 0 \leq s_k \leq 1, \quad (30)$$

the estimation error obeys

$$E_{k+1} \leq (1 - \gamma)E_k + \gamma\{(1 - s_k)\|f_k\| + s_k \epsilon_k\} + \nu_k. \quad (31)$$

For constant noiseless observations and constant s , the unique fixed point is $\Pi_C(sf)$, approached with contraction factor at most $1 - \gamma$. If $sf \in C$, the fixed point is sf , which is biased unless $s = 1$ or $f = 0$.

Proof. Apply projection nonexpansiveness relative to f_k and use $s_k z_k - f_k = -(1 - s_k)f_k + s_k w_k$ to obtain the bound. For constant target sf , the update map is a contraction with factor $1 - \gamma$. Let $p = \Pi_C(sf)$. The characterization $\langle sf - p, x - p \rangle \leq 0$ for all $x \in C$ also implies $\Pi_C\{p + \gamma(sf - p)\} = p$. Thus p is its unique fixed point. $\square$

The documented observation-attenuated normalization has

$$s_k = \frac{\mathbf{1}[\|P_r e_k^{\text{res}}\| > \delta]}{1 + \|P_r e_k^{\text{res}}\|^2 / \rho^2},$$

subject to the selected-channel norm and calibration convention. Its guaranteed error includes attenuation. The innovation proposition does not retrospectively give an unbiased-estimation guarantee to experiments run with this law.

Update and hold are distinct from target-only gating. Let k index every physical command, including steps with no estimator update. The associated measurement becomes available after executing u_k and can first affect $\hat{f}_{k+1}$. A full-update indicator $\chi_k \in \{0, 1\}$ gives the exact scheduled maps

$$\hat{f}_{k+1} = \Pi_C\{(1 - \chi_k \gamma)\hat{f}_k + \chi_k \gamma s_k z_k\}, \quad (32)$$

$$\hat{f}_{k+1} = \Pi_C\{\hat{f}_k + \chi_k \alpha_k (z_k - \hat{f}_k)\}. \quad (33)$$

Because $\hat{f}_k \in C$, $\chi_k = 0$ holds the estimate exactly. Applying the two projection arguments above with the effective step sizes $\chi_k \gamma$ or $\chi_k \alpha_k$ proves the full-error counterparts of Eqs. (7)–(8). In particular, a full hold has $E_{k+1} \leq E_k + \nu_k$. In contrast, zeroing a legacy observation while retaining leakage uses $\chi_k = 1$, $s_k = 0$ and gives $E_{k+1} \leq (1 - \gamma)E_k + \gamma\|f_k\| + \nu_k$. This can move the estimate toward zero even when the physical fault persists. The schedule indicator must represent the executed code path; it is not inferred from the displayed deadzone threshold.

Both recursions have the form

$$B_{k+1} = (1 - \beta_k)B_k + \beta_k\omega_k + \nu_k, \quad B_0 \geq E_0, \quad (34)$$

where (β_k, ω_k) equals $(\chi_k \gamma, (1 - s_k)\|f_k\| + s_k \epsilon_k)$ for observation attenuation and $(\chi_k \alpha_k, \epsilon_k)$ for innovation adaptation. With $P_{j,k}^\beta = \prod_{i=j}^{k-1} (1 - \beta_i)$,

$$E_k \leq B_k = P_{0,k}^\beta B_0 + \sum_{j=0}^{k-1} P_{j+1,k}^\beta (\beta_j \omega_j + \nu_j). \quad (35)$$

This follows by induction and is valid for arbitrary finite schedules, including long holds and a fault jump represented by a single large ν_j . For constant s and noiseless constant f , the distance to the legacy fixed point $\Pi_C(sf)$ is at most $(1 - \gamma)^{N_k}$ times its initial value, where $N_k = \sum_{j=0}^{k-1} \chi_j$. It need not decay at the same rate per physical step. Converting the count into seconds requires actual timestamps; a positive ungated gain alone supplies no such rate.

Selected-coordinate proof of Proposition 1. For a box $C = \prod_i [\ell_i, u_i]$ and binary diagonal D , coordinatewise clipping gives, for any v and $f \in C$,

$$\|D(\Pi_C(v) - f)\|^2 = \sum_{i:D_{ii}=1} |\Pi_{[\ell_i, u_i]}(v_i) - f_i|^2 \leq \|D(v - f)\|^2. \quad (36)$$

Apply this inequality to each scheduled update, expand $z_k = f_k + w_k$, and add $\|D(f_{k+1} - f_k)\|$. This proves the two main masked recursions. The scalar envelope and product in Eqs. (34)–(35) then hold with B_k^D, E_k^D, ν_k^D and $\omega_k^D = (1 - s_k)\|Df_k\| + s_k \epsilon_k^D$ for attenuation, or $\omega_k^D = \epsilon_k^D$ for innovation.

This conclusion requires separable projection. For a counterexample with a coupled set, take the unit disk, $f = (1/2, 0)$, $v = (1/2, 2)$, and $D = \text{diag}(1, 0)$. Then $D(v - f) = 0$, whereas the first coordinate of the projection is $1/\sqrt{17} \neq 1/2$. Thus selected error can increase through projection even when the selected input error is zero. The full-error proof for arbitrary closed convex sets remains valid.

Application snapshots and buffered commands. Let $\hat{f}_k^{\text{app}} = \hat{f}_{\tau_k}$ be the estimate used in physical command k , $0 \leq \tau_k \leq k$. Define $E_k^{D, \text{app}} = \|D(\hat{f}_{\tau_k} - f_k)\|$. Inserting respectively $\hat{f}_k$ and f_{τ_k} gives

$$E_k^{D, \text{app}} \leq E_k^D + \sum_{j=\tau_k}^{k-1} \|D(\hat{f}_{j+1} - \hat{f}_j)\|, \quad (37)$$

$$E_k^{D, \text{app}} \leq E_{\tau_k}^D + \sum_{j=\tau_k}^{k-1} \nu_j^D. \quad (38)$$

Taking the minimum of these valid bounds proves Eq. (11). For maximum age d and masked drift ν , the second is $B_{\tau_k}^D + d\nu$. An episode-long hold uses its full age. No delay limit is needed for the finite-horizon forms. Setting $D = I$ yields the full-error bounds $E_k^{\text{app}} \leq B_k^{\text{app}}$ used for a general convex projection or nonbinary correction map. Snapshot creation time, full holds, and target-only gates are distinct quantities. If one decoder chunk uses multiple estimates, its commands must be represented at the matching finer resolution, or the discrepancy must be included in ζ_k .

B.3 WHAT CONTRACTS IN THE IMPLEMENTED AND CANDIDATE OBSERVERS

The innovation bound controls error to truth along the realized gain sequence. It does not omit the gain derivative from a pairwise Jacobian. For scalar fixed data z and inactive projection, put $e = z - h$

and $v = e^2/\rho^2$. Differentiating the actual map gives

$$\frac{d}{dh} \left(h + \frac{\gamma e}{1+v} \right) = 1 - \frac{\gamma}{1+v} + \frac{2\gamma v}{(1+v)^2} = 1 + \gamma \frac{v-1}{(1+v)^2}.$$

The derivative of $(v-1)/(1+v)^2$ is $(3-v)/(1+v)^3$, proving the maximum at $v = 3$ in Eq. (9). When $z = f$, the truth error is still multiplied by $1 - \gamma/(1+v)$. This interior counterexample rules out global Euclidean contraction of this map; it does not rule out another metric on a restricted domain. Conversely, the legacy map conditionally contracts under a common fixed observation stream but can converge to the biased attenuated target. Neither property certifies contraction of the robot–estimator feedback interconnection.

Proposition 6 (A different sampled prediction-gradient candidate). *Let Θ be closed and convex, $\Gamma \succ 0$ fixed, and $H_{I,k} = H_{I,k}^\top \succeq 0$. Condition on a common exogenous data stream $(H_{I,k}, b_{I,k})$ and gains h_k independent of the compared estimates. The weighted prediction loss has gradient $H_{I,k}\theta - b_{I,k}$; its projected negative-gradient step is*

$$T_k(\theta) = \Pi_{\Theta}^{\Gamma^{-1}} [\theta + h_k \Gamma (b_{I,k} - H_{I,k}\theta)]. \quad (39)$$

Define $S_{I,k} = \Gamma^{1/2} H_{I,k} \Gamma^{1/2}$ and $q_{\text{obs},k} = \max_i |1 - h_k \lambda_i(S_{I,k})|$. Then

$$\|T_k(\theta_1) - T_k(\theta_2)\|_{\Gamma^{-1}} \leq q_{\text{obs},k} \|\theta_1 - \theta_2\|_{\Gamma^{-1}}. \quad (40)$$

For $S_{I,k} \succ 0$, $0 < h_k < 2/\lambda_{\max}(S_{I,k})$ gives strict conditional contraction. A full hold has $h_k = 0$ and factor one.

Proof. Under $v = \Gamma^{-1/2}\theta$, metric projection is Euclidean projection onto $\Gamma^{-1/2}\Theta$, hence nonexpansive even at active faces. The unprojected difference is $(I - h_k S_{I,k})(v_1 - v_2)$; its symmetric-matrix operator norm is exactly $q_{\text{obs},k}$. This proves the bound without differentiating through projection. The step-size condition is equivalent to $|1 - h_k \lambda_i| < 1$ for every positive eigenvalue. $\square$

If $b_{I,k} = H_{I,k}\theta_k^* + \epsilon_{I,k}$ and $\theta_k^* \in \Theta$, the same argument gives the truth-error bound

$$Z_{k+1} \leq q_{\text{obs},k} Z_k + h_k \|\Gamma^{1/2} \epsilon_{I,k}\| + \|\theta_{k+1}^* - \theta_k^*\|_{\Gamma^{-1}}, \quad (41)$$

for $h_k \geq 0$. Uniform exponential forgetting requires a uniform factor below one or a proved windowed product condition. Rank-deficient information precludes per-step strict contraction in every parameter direction. A FIR command buffer alone is not an informative parameter dataset. Additive $z_k = f_k + w_k$ with already known invertible sensitivity corresponds to $H_I = I$, $b_I = z$, $\Gamma = I$; no new command excitation is then needed. Unknown gains and offsets require separate regressor-rank conditions.

Equation (39) is an unevaluated candidate, not a replacement interpretation of the reported attenuation or state-dependent innovation runs. A gain selected from data quality or regressor energy is admissible here only after conditioning on a common stream that does not depend on the compared estimates. The actual robot stream changes when commands change; the coupled analysis below addresses that separate issue.

B.4 FROM AN EXECUTION METRIC TO A SCHEDULED RECOVERY TUBE

Execution state, command source, and observer state. The certified state ξ_k contains the physical robot and necessary low-level controller state, including actuator delays, filters, and reference interpolation memory when required for a closed execution model. Manipulation also requires the physical-state closure specified below. The VLA supplies a_k ; its observation history, network state, and policy memory belong to the command source and are not part of ξ_k . The observer’s issued pre-fault command history, estimate, gains, and applied snapshot belong to a separate state ξ_k^{obs} . A received nominal chunk may be treated as an external command schedule; if its interpolation or phase is handled inside the execution controller, the required controller memory must instead be represented in ξ_k . The chosen realization and command schedule must be explicit.

Physical state closure during manipulation. If object pose, velocity, deformation, or evolving contact variables affect execution, include their necessary state in ξ_k and its nominal comparator, together with the relevant contact mode. A discrete mode indexes the vector field and metric; it is not a smooth Euclidean coordinate. Fixed environment geometry may be a common model parameter; prescribed environment motion may be a declared common exogenous input. An object moved by the robot is an endogenous physical state, not a common input merely because the commands are shared. The metric and its domains must cover the selected physical state and all allowed geometry, modes, and transitions. The actual and nominal systems receive the same commands but can reach different object states and contact sequences; agreement of those sequences is not assumed by notation.

Alternatively, a reduced servo state may omit these variables if their remaining influence admits an explicit uniform transition-error bound η_k^{contact} as specified below. This is a bound over admissible corrected states, commands, omitted physical states, and contact events; it need not be small or vanish under perfect actuator correction. Without full state closure or such a bound, a servo/free-space certificate does not extend to contact manipulation. Even a complete physical state does not by itself establish contraction of its dynamics.

The sufficient closure condition for a q_k -only execution transition is: at fixed ξ_k and supplied command, any two observer/adaptor states producing the same realized effective mismatch q_k must produce the same next execution state. With no separate transition disturbance, $q_k = 0$ then recovers the nominal execution map. A corrected-command queue may be outside ξ_k only if its entire effect is mediated by the command actually executed; plant or controller delays cannot simply be discarded. Under this closure, condition on the realized supplied commands and write

$$\xi_{k+1}^{\text{nom}} = F_{\text{exec},k}^0(\xi_k^{\text{nom}}, a_k), \quad \xi_{k+1} = F_{\text{exec},k}(\xi_k, a_k, q_k). \quad (42)$$

The reference is the nominal physical response to these same commands. It does not rerun the VLA on hypothetical healthy observations. The commands may be generated using the actual robot's observations: the result is then a pathwise execution comparison, valid while those commands and the states remain in the assumed admissible sets. It does not establish stability of the feedback law that generated the commands. Computing ξ_k^{nom} online is unnecessary for the evaluated prediction-only laws.

For binary D and box projection,

$$\|q_k\| \leq U_k^\perp + E_k^{D,\text{app}} + \bar{\zeta}_k, \quad U_k^\perp = \|(I - D)f_k\|. \quad (43)$$

The supplied nominal interface action a_k is the one used to define the actual mismatch; it is not a newly queried healthy-policy action. For general D or a coupled projection use instead $\|q_k\| \leq U_k^\perp + \|D\|B_k^{\text{app}} + \bar{\zeta}_k$. Native-decoder error and command clipping belong in ζ_k ; any other separate transition disturbance belongs in η_k below. Each physical effect is counted once. In particular, bounded parameter projection does not imply command feasibility or repair of unmatched joint faults.

Additional history and contact effects. If a plant or low-level controller needs a history not determined by (ξ_k, a_k, q_k) , retain the required memory in ξ_k and its nominal comparator, or use an effective-input history with an exact realization. A VLA policy that reads raw issued-action history changes the supplied command stream; this dependence alone does not place its memory inside the primary execution certificate. If a raw-command or corrected-queue update has an additional direct effect on the certified execution state, write its explicit map as $\mathcal{F}_{\text{exec},k}(\xi, a, q, h)$ with $\mathcal{F}_{\text{exec},k}(\xi, a, q, 0) = F_{\text{exec},k}(\xi, a, q)$. Require, uniformly on the certified domain and over allowed commands,

$$\begin{aligned} d_{k+1}(\mathcal{F}_{\text{exec},k}(\xi, a, q, h), F_{\text{exec},k}(\xi, a, q)) &\leq L_k^h \|h\|, \\ \|h_k\| &\leq \bar{h}_k, \\ d_{k+1}(\xi_{k+1}, \mathcal{F}_{\text{exec},k}(\xi_k, a_k, q_k, h_k)) &\leq \eta_k^{\text{contact}}, \\ \eta_k &= \eta_k^{\text{base}} + \eta_k^{\text{hist}} + \eta_k^{\text{contact}}, \quad \eta_k^{\text{hist}} = L_k^h \bar{h}_k. \end{aligned} \quad (44)$$

The third line bounds the remaining omitted contact/environment effect on the reduced-state transition; it is zero when that effect is already represented exactly. Its validity must include the relevant event timing and resets. Here η_k^{base} bounds other base-map disturbance, excluding the displayed history/contact effects and realization error already included in q_k . The triangle inequality gives

$$\begin{aligned} d_{k+1}(\xi_{k+1}, F_{\text{exec},k}^0(\xi_k, a_k)) &\leq L_k \|q_k\| + \eta_k^{\text{base}} + L_k^h \|h_k\| + \eta_k^{\text{contact}} \\ &\leq L_k \|q_k\| + \eta_k. \end{aligned}$$

These are required bounds, not definitions that make omitted effects small or make them vanish at $q_k = 0$. In a reduced model, the residual bound is required for every admissible hidden-state realization. A different execution-state realization requires its own distance-contraction and transition check. The tube proof uses total forcing η_k , with each physical effect counted once.

For example, take $p_{k+1} = .8p_k + .2(u_k + f)$, supplied nominal $a_k = 0$, a constant nonzero fault f , and exact correction $u_k = -\hat{f}_k = -f$. Then $q_k = 0$ and physical motion matches $p_{k+1}^{\text{nom}} = .8p_k^{\text{nom}}$ from equal initial positions. Nevertheless, a raw-command memory $v_{k+1} = u_k$ equals $-f$, whereas $v_{k+1}^{\text{nom}} = 0$. Starting with equal (p_0, v_0) , their next augmented distance is $|f|$, so a zero-forcing q -only tube for (p, v) fails even though its nominal Euclidean map contracts by $.8$. For the physical execution state $\xi = p$, the q -only model is exact; v can stay in the observer. If (p, v) is certified instead, its exact augmented map $\mathcal{F}((p, v), q, h) = (.8p + .2q, h)$ with $h = u - a$ has $L = .2$, $L^h = 1$, and $\bar{h} = |f|$ under exact correction. The conditional tube proof is unchanged; its state and forcing must represent the chosen subsystem.

Direct contraction and smooth sufficient conditions. Theorem 2 assumes the distance contraction in Eq. (13) directly. For a smooth-mode sufficient construction, let $M_{c,k}(\xi)$ be smooth, independent of the supplied command and adapted estimate, and satisfy $0 < \underline{m}I \preceq M_{c,k} \preceq \bar{m}I$ on the stated domains; let d_k be its intrinsic Riemannian distance. With Euclidean δ_{phys} these bounds imply Eq. (12) with $c_- = \sqrt{\underline{m}}$ and $c_+ = \sqrt{\bar{m}}$. Suppose $F_{\text{exec},k}^0(\cdot, a)$ is C^1 and, uniformly over allowed a ,

$$J_k^\top M_{c,k+1}(F_{\text{exec},k}^0(\xi, a)) J_k \preceq (\lambda_k^{\text{exec}})^2 M_{c,k}(\xi), \quad J_k = \partial_\xi F_{\text{exec},k}^0(\xi, a), \quad (45)$$

with a held fixed. If admissible connecting curves have admissible nominal images, this inequality contracts each image-curve length by λ_k^{exec} ; taking the infimum proves Eq. (13). Thus Eq. (45) is sufficient in a smooth mode but is not required when Eq. (13) is established directly. A geodesically convex neighborhood with its relevant images contained in the next domain is sufficient for this construction; no VLA Jacobian appears.

Separately, Theorem 2 assumes the realized-transition bound in Eq. (14). Under this Riemannian construction, if $F_{\text{exec},k}$ is differentiable in its actual physical input, a sufficient input-sensitivity condition is

$$\left\| M_{c,k+1}^{1/2}(F_{\text{exec},k}(\xi, a, v)) D_q F_{\text{exec},k}(\xi, a, v) \right\| \leq L_k \quad \text{for every } v = sq, 0 \leq s \leq 1.$$

Integrating the length of $s \mapsto F_{\text{exec},k}(\xi, a, sq)$, then adding $d_{k+1}(F_{\text{exec},k}(\xi, a, 0), F_{\text{exec},k}^0(\xi, a)) \leq \eta_k$, proves

$$d_{k+1}(F_{\text{exec},k}(\xi, a, q), F_{\text{exec},k}^0(\xi, a)) \leq L_k \|q\| + \eta_k. \quad (46)$$

For the explicit decomposition in (44), the base-map argument uses η_k^{base} instead; adding its history and contact bounds once gives the same inequality for the realized next state and hence Eq. (14). The entire input path must lie in the domain. For a nonsmooth map, establish Eq. (14) directly. The condition concerns the realized physical input, not merely the pre-decoder correction parameter.

Proof and quantitative form of Theorem 2. Insert $F_{\text{exec},k}^0(\xi_k, a_k)$ between the two next states and apply the triangle inequality, Eqs. (13) and (14), and the masked mismatch bound. Thus $X_{k+1} \leq \lambda_k^{\text{exec}} X_k + g_k$, with $X_k = d_k(\xi_k, \xi_k^{\text{nom}})$, where

$$g_k = L_k(U_k^\perp + B_k^{D,\text{app}} + \bar{\zeta}_k) + \eta_k, \quad R_{k+1} = \lambda_k^{\text{exec}} R_k + g_k. \quad (47)$$

Induction proves inclusion from $R_0 \geq d_0(\xi_0, \xi_0^{\text{nom}})$ as long as the radius- R_k balls remain in the domains. Expansion gives

$$R_k = \left(\prod_{i=0}^{k-1} \lambda_i^{\text{exec}} \right) R_0 + \sum_{j=0}^{k-1} \left(\prod_{i=j+1}^{k-1} \lambda_i^{\text{exec}} \right) g_j. \quad (48)$$

Arbitrary finite nonnegative amplification factors give only this finite-horizon statement. If every subinterval satisfies $\prod_{i=j}^{k-1} \lambda_i^{\text{exec}} \leq C \varrho^{k-j}$, $C \geq 1$, $0 < \varrho < 1$, then

$$R_k \leq C \varrho^k R_0 + C \sum_{j=0}^{k-1} \varrho^{k-1-j} g_j \leq C \varrho^k R_0 + \frac{C \bar{g}}{1 - \varrho} (1 - \varrho^k) \quad (49)$$

when $g_j \leq \bar{g}$. This allows transient expansion; a uniform $\lambda_k^{\text{exec}} \leq a < 1$ is the special case $C = 1, \varrho = a$. If $0 \leq a, q < 1$, $\lambda_k^{\text{exec}} \leq a$, $L_k \leq L$, $U_k^\perp \leq U^\perp$, $\bar{\zeta}_k \leq \bar{\zeta}$, $\eta_k \leq \eta$, and $B_k^{D, \text{app}} \leq q^k E_0^D + \bar{E}$, then

$$R_k \leq a^k R_0 + L E_0^D C_k(a, q) + \frac{L(U^\perp + \bar{E} + \bar{\zeta}) + \eta}{1 - a} (1 - a^k), \quad (50)$$

where $C_k(a, q) = \sum_{j=0}^{k-1} a^{k-1-j} q^j$ equals $(a^k - q^k)/(a - q)$ if $a \neq q$, and ka^{k-1} if $a = q$ for $k \geq 1$; $C_0 = 0$. Bounds on the applied error must already include its actual schedule; estimate convergence alone does not imply this geometric form.

Hybrid execution, chunks, and reference jumps. At a step containing an impact, reset, or mode change, Eq. (13) must hold for the complete nominal sampled transition; a within-mode Jacobian check alone is insufficient. If the domains form a hybrid union, d_k must be defined for the cross-mode state pairs used by the comparison, or the certified event structure must exclude those pairs. The physical certificate includes relevant controller memory, actuator delays, and contact modes. Within a smooth mode, flow inequalities apply only on its certified domain. Impulsive contact forces are not ordinary bounded continuous disturbances: impacts require state/metric jump bounds, or a verified sampled transition bound that includes the impact. State-dependent event timing requires a valid saltation or incremental event-map bound; a reset Jacobian alone can omit event-time sensitivity. The compared executions need not impact at the same time. A common-mode flow estimate alone does not compare differing mode sequences; either justify the required hybrid comparison, restrict the admissible events, or bound the resulting sampled transition discrepancy explicitly. Uniform bounds can fail near grazing events or changes in contact sequence. A chunk-boundary metric may be appropriate when buffer shifts obstruct one-step contraction, but intermediate physical motion requires its own bounds. Boundary stability does not certify intermediate constraints.

Both physical systems receive the same command changes. In physical-state coordinates their responses need not jump when the command changes; no command derivative enters a command-independent metric. A commanded reset of low-level controller state must nevertheless be included in the transition. If tracking coordinates are instead measured directly from a commanded reference that jumps, retain the resulting coordinate/reset jump and its metric gain or additive forcing, or use a feasible interpolated reference. Common commands cannot justify omitting a reset that the chosen coordinates actually undergo. This conditional pathwise result is neither a distributional success theorem nor a certificate of the VLA command source’s feedback stability.

Optional conditional corollary: task preservation. Let $\mathcal{S}$ be the successful physical trajectories over horizon N , including any specified command-dependent constraints. Suppose the nominal execution trajectory $\xi_{0:N}^{\text{nom}}$ for the supplied command sequence is successful and has positive radii μ_k^{task} such that every admissible trajectory with $d_k(\xi_k, \xi_k^{\text{nom}}) < \mu_k^{\text{task}}$ at all steps belongs to $\mathcal{S}$. If the execution-tube assumptions hold and $R_k < \mu_k^{\text{task}}$ throughout, the realized execution succeeds. This follows immediately from tube inclusion and the assumed robustness neighborhood. The neighborhood must cover all path and terminal conditions and intermediate chunk motion; constraints on omitted variables require separate bounds. Neither success of the supplied nominal response nor its whole-task margin follows from contraction.

When comparing two adapters, their observation feedback may produce different VLA command streams. Fixed-command replay gives the common-input comparison directly; otherwise each adapter first has its own nominal physical response. In a common certified metric, the triangle inequality then bounds their physical separation by the sum of their two execution radii and the distance between those nominal responses. That last command-divergence term requires separate evidence and cannot be set to zero because the VLA weights are frozen.

An independently rerun healthy VLA generally supplies different commands. Comparing against that rollout requires a separate bound on the divergence between its nominal physical response and ξ_k^{nom} . For example, if such a bound is $d_k(\xi_k^{\text{nom}}, \xi_k^h) \leq C_k^{\text{cmd}}$, the triangle inequality gives $d_k(\xi_k, \xi_k^h) \leq R_k + C_k^{\text{cmd}}$; a task margin around ξ_k^h must exceed this sum. A bound on command differences together with a valid execution input-sensitivity bound can provide C_k^{cmd} , but it must be established separately. Neither a VLA metric nor this outer-loop comparison is needed for the primary execution tube. The reported experiments establish no task margins. A uniform physical

execution metric was measured directly by same-command replay rather than inferred from fitted gains or sampled Jacobians, and exists on the GR1 arm but not on the Panda under operational-space control (Appendix G.15).

B.5 CLOSING THE LOOP WHEN PREDICTOR ERROR DEPENDS ON TRAJECTORY ERROR

An independently bounded ϵ_k can be a strong assumption: a correction changes the visited physical states and commands, and predictor error can grow with execution deviation and estimation error. The assumed error bound must hold uniformly over admissible supplied commands; it does not follow from a healthy-data fit or from the physical metric alone. The following result gives a sufficient condition for closing this feedback loop. It applies when every physical step updates and applies the current estimate, with a uniformly positive active step size. Observation attenuation is included by adding its forcing to the constant observation-error term below. Arbitrary holds and delayed application instead use the scheduled tube.

Proposition 7 (A conditional small-gain bound). *Let $X_k = d_k(\xi_k, \xi_k^{\text{nom}})$ be the same-command execution distance and write $E_k = E_k^D$ within this subsection. Suppose on the relevant local domain*

$$\begin{aligned} X_{k+1} &\leq \lambda X_k + bE_k + d_0, \\ E_{k+1} &\leq [1 - \alpha_k(1 - k_E)]E_k + \alpha_k(\epsilon_0 + k_X X_k) + \nu, \end{aligned} \quad (51)$$

where $0 \leq \lambda < 1$, $0 \leq k_E < 1$, $b, k_X, \epsilon_0, \nu, d_0 \geq 0$ and $0 < \underline{\alpha} \leq \alpha_k \leq \bar{\alpha} \leq 1$. The first inequality follows from the tube assumptions with $b = L$ and $d_0 = L(U^\perp + \zeta) + \eta$; equivalently, an adapter discrepancy can be absorbed into η_k as $L_k \bar{\zeta}_k$, but must not be counted twice. For innovation adaptation the second follows from $\|Dw_k\| \leq \epsilon_0 + k_E E_k + k_X X_k$. For the observation-attenuated law take $\alpha_k = \gamma$ and replace ϵ_0 throughout this proposition by $\epsilon_0 + \bar{a}_{\text{att}}$. If

$$k_X b < (1 - k_E)(1 - \lambda), \quad (52)$$

choose $s > 0$ with $b/(1 - \lambda) < s < (1 - k_E)/k_X$, with the upper bound interpreted as $+\infty$ when $k_X = 0$. Then $W_k = \max\{X_k, sE_k\}$ satisfies

$$W_k \leq \varrho^k W_0 + \frac{d_*}{1 - \varrho} (1 - \varrho^k), \quad (53)$$

where

$$\varrho = \max\{\lambda + b/s, 1 - \underline{\alpha}(1 - k_E - sk_X)\} < 1, \quad d_* = \max\{d_0, s(\bar{\alpha}\epsilon_0 + \nu)\}. \quad (54)$$

The bound holds inductively as long as its implied trajectory and estimator neighborhoods remain inside the domain of the assumptions.

Proof. The strict small-gain inequality makes the interval for s nonempty. Since $X_k \leq W_k$ and $E_k \leq W_k/s$, the first recursion gives $X_{k+1} \leq (\lambda + b/s)W_k + d_0$. The second gives

$$\begin{aligned} sE_{k+1} &\leq [1 - \alpha_k(1 - k_E) + sk_X \alpha_k]W_k + s(\alpha_k \epsilon_0 + \nu) \\ &\leq [1 - \underline{\alpha}(1 - k_E - sk_X)]W_k + s(\bar{\alpha}\epsilon_0 + \nu). \end{aligned}$$

Both coefficients are strictly below one. Taking the maximum and expanding $W_{k+1} \leq \varrho W_k + d_*$ proves the claim. $\square$

For a constant step size, the homogeneous comparison matrix is

$$A_\alpha = \begin{pmatrix} \lambda & b \\ \alpha k_X & 1 - \alpha(1 - k_E) \end{pmatrix}.$$

The weighted-maximum proof also covers the stated time-varying family. The result is a stability bound for the coupled error comparison, not a claim of global pairwise contraction of the nonlinear estimator. It explains why a well-conditioned sensitivity and small predictor error growth can matter even when healthy prediction fits appear good. The constants k_E, k_X, λ were measured on the deployed plant (Appendix G.15): the coupling terms are small ($k_E = .012$, $k_X = .009$, $\epsilon_0 = .0008$), and the condition nonetheless fails because $a = 1 - \lambda = 9 \cdot 10^{-5}$. The obstruction is the absent contraction, not the feedback coupling this subsection bounds.

B.6 INVARIANT RECOVERY REGIONS AND ADMISSIBLE STARTUP ERROR

Proof of Corollary 3. Throughout this subsection, E_k abbreviates E_k^D , and ν bounds masked drift. In the active-every-step case with current application, the two estimator recursions imply the common comparison

$$E_{k+1} \leq [1 - \beta_k c] E_k + \beta_k (k_X X_k + v),$$

where $\beta_k = \gamma$ for observation attenuation and $\beta_k = \alpha_k$ for innovation adaptation. For the first case use $(1 - s_k) \|Df_k\| \leq \bar{a}_{\text{att}}$ and $s_k \leq 1$; for the second use $\nu \leq \alpha_k \nu / \alpha$. The coefficients multiplying X_k, E_k are nonnegative because $0 \leq k_E < 1$ and $0 < \beta_k \leq 1$. At the upper corner of the rectangle, its second inequality gives $\beta_k (k_X \bar{X} + v) \leq \beta_k c \bar{E}$. Hence $E_{k+1} \leq \bar{E}$. The first inequality gives $X_{k+1} \leq \bar{X}$. Induction from the stated initial conditions proves invariance, and the assumed domain containment makes each application valid. Solving the two corner equalities gives the equality corner

$$X_* = \frac{cd_0 + bv}{\Delta}, \quad E_* = \frac{k_X d_0 + av}{\Delta}. \quad (55)$$

A task-preservation conclusion additionally requires the successful nominal response and whole-trajectory neighborhood in Appendix B.4. $\square$

For a chosen $\bar{X} < \mu_{\text{exec}}$, where μ_{exec} is a valid execution-domain tube radius, and $b > 0$, a sufficient startup budget given valid parameter-error and domain bounds is

$$bv + cd_0 \leq \Delta \bar{X}, \quad bE_0 + d_0 \leq a\bar{X}, \quad X_0 \leq \bar{X}, \quad (56)$$

with estimator radius chosen from the nonempty interval

$$\max\{E_0, (k_X \bar{X} + v)/c\} \leq \bar{E} \leq (a\bar{X} - d_0)/b. \quad (57)$$

The second condition in Eq. (56) is essential: the steady-state budget alone does not accommodate an arbitrarily large fault-onset estimation error. For $b = 0$, use the two original rectangle inequalities directly. A common scaling of (X_*, E_*) by a factor at least one preserves both inequalities. It can cover arbitrary finite initial errors when both corner components are positive. If either is zero, use the original rectangle inequalities or the product bound directly.

For completeness, the primary observation-attenuated law has the explicit comparison below. Suppose $\bar{a}_{\text{att}} \geq (1 - s_k) \|Df_k\|$ and $\|Dw_k\| \leq \epsilon_0 + k_E E_k + k_X X_k$. Since $0 \leq s_k \leq 1$, the masked counterpart of Eq. (31) implies

$$E_{k+1} \leq [1 - \gamma(1 - k_E)] E_k + \gamma(\epsilon_0 + \bar{a}_{\text{att}} + k_X X_k) + \nu. \quad (58)$$

The same rectangle proof applies with $v = \epsilon_0 + \bar{a}_{\text{att}} + \nu/\gamma$. It is conservative; retaining s_k in the original recursion can be tighter. The attenuation floor cannot be removed when this is the executed law.

An exact step-history contribution. For an otherwise constant fault jumping by Δf at k_0 , the history part of Eq. (22) at $j = k - k_0 \geq 0$ is

$$w_{k_0+j}^{\text{hist}} = -S^{-1} \sum_{\ell=j+1}^K G_\ell \Delta f. \quad (59)$$

For each term with $\ell > j$, the delayed fault precedes the jump and $f_{k-\ell} - f_k = -\Delta f$; every other term is zero. This proves the identity and its disappearance after K steps. Substituting its time-dependent norm into the estimator and trajectory recursions accounts for onset without treating the jump as a persistent drift rate.

Rate, noise, and drift are separate quantities. For fixed s , constant noiseless observations, and inactive projection, the legacy transient is $\hat{f}_k - sf = (1 - \gamma)^k (\hat{f}_0 - sf)$. With $\gamma = .08$, 36 updates reduce that transient by at least 95%; this concerns the attenuated target. For a constant-gain innovation observer with independent zero-mean scalar measurement noise of variance σ_w^2 , inactive projection, and constant fault, the stationary error variance is $\alpha \sigma_w^2 / (2 - \alpha)$, obtained by solving $v = (1 - \alpha)^2 v + \alpha^2 \sigma_w^2$. This calculation is illustrative: correlated FIR residuals, normalization, and gates require their own stochastic analysis. The deterministic drift bound remains ν/α .

B.7 WHY IDENTIFYING FIRST AND THEN HOLDING CHANGES THE BOUND

Suppose an estimate obtained before an evaluation episode is held at $\hat{f}^H$, with $\|D(f_0 - \hat{f}^H)\| \leq E_H$. If the persistent fault can drift in selected coordinates by at most ν per control step, then

$$\|D(f_k - \hat{f}^H)\| \leq E_H + k\nu, \quad (60)$$

by telescoping. Substituting this quantity for $B_k^{D,\text{app}}$ in (47) yields the held-estimate guarantee. For a constant fault and a valid initialization, holding removes the online estimation transient from the task episode, but retains calibration error; for a changing fault it accumulates drift. A prior identification episode also changes the information available at evaluation time. Its benefit cannot be attributed solely to removing within-episode variance or compared as if both protocols started with zero fault information.

B.8 GAIN FAULTS AND ADAPTER DIMENSION

For a diagonal per-channel FIR and a gain g_i constant over the FIR window, the correct scalar regression is

$$e_{i,k}^{\text{res}} = \beta_i \phi_{i,k} + \nu_{i,k}, \quad \beta_i = g_i - 1, \quad \phi_{i,k} = \sum_{\ell=0}^K h_{i,\ell} u_{i,k-\ell} = \hat{y}_{i,k} - c_i. \quad (61)$$

The equality is exact only for the stated diagonal, correctly calibrated plant; coupling and time-varying gains introduce model error. In particular, gain drift contributes $\sum_{\ell} h_{i,\ell} (g_{i,k-\ell} - g_{i,k}) u_{i,k-\ell}$. Noiseless constant-gain identification requires $\sum_k \phi_{i,k}^2 > 0$; robustness depends on its magnitude. With an unknown constant intercept, the two-column regression $[\phi_{i,k}, 1]$ has full rank only when $\phi_{i,k}$ varies. Command quantile ranges alone establish neither of these conditions.

The inverse-gain adapter $u_i = a_i / \hat{g}_i$, with $\hat{g}_i \geq g_{\min} > 0$, gives

$$q_i = \left(\frac{g_i}{\hat{g}_i} - 1 \right) a_i = \frac{g_i - \hat{g}_i}{\hat{g}_i} a_i, \quad |q_i| \leq \frac{|a_i|}{g_{\min}} |g_i - \hat{g}_i|. \quad (62)$$

This is the physical mismatch to insert in the trajectory tube. The derivative of the *command* with respect to its estimate is $-a_i / \hat{g}_i^2$; the derivative of the *executed command* is $-g_i a_i / \hat{g}_i^2$. They have different amplification factors. Input limits must still permit the required inverse command.

Proposition 8 (Local correction authority and dimension). *Suppose a family of constant faults is $f = V\theta$ and an additive adapter can generate $B\vartheta$ with fixed $B \in \mathbb{R}^{d \times p}$. Exact action-level compensation of the whole family by unrestricted adapter coordinates is possible if and only if*

$$\text{Range}(V) \subseteq \text{Range}(B), \quad \text{which implies} \quad p \geq \text{rank}(V). \quad (63)$$

Proof. For necessity, apply the matching requirement to each coordinate basis vector θ ; every column of V must then lie in the range of B . For sufficiency, choose K with $BK = V$ and set $\vartheta = K\theta$ (with the compensation sign in the commanded action). $\square$

With bounded adapter coordinates, the required solutions must also be feasible inside those bounds; range inclusion alone does not ensure that. If only a local physical effect matters, the corresponding criterion uses $\text{Range}(AV) \subseteq \text{Range}(AB)$ for the linearized action-to-effect map A . For one shared parameter vector across multiple times, the time-indexed maps must be stacked before applying this range criterion. A six-channel interface spans arbitrary six-dimensional offsets, whereas a known uniform offset $f = \alpha \mathbf{1}$ is a one-dimensional family. The experiments do not establish that six is a universal minimum. A joint lock can violate local matching or command feasibility; it does not prove that another policy or a different path through the remaining joints could never complete the task.

C STRUCTURED PHYSICAL OOD AND THE CONTRACTION INTERCONNECTION

This appendix specifies the model and operating conditions behind Appendix D, then derives the prediction-only comparison. The general composite controller is a theoretical extension; the reported VLA runs retain the sampled laws in Section 2.

C.1 PHYSICAL MATCHING AND THE SAME-COMMAND REFERENCE

The frozen VLA supplies a command signal $r(t)$, which can be piecewise continuous with locally finitely many jumps and can depend causally on deployment observations. Fix that entire supplied signal when defining the execution comparison. The state ξ contains the robot and necessary low-level controller state; $k_0(\xi, r, t)$ is its nominal execution feedback. For manipulation, include evolving object and contact/environment state needed to close these dynamics, or explicitly bound its omitted influence below. Static geometry can be a fixed common parameter. Endogenous object motion is not a common input merely because both executions receive the same commands (Appendix B.4). Write

$$\dot{\xi} = f_0(\xi, t) + G(\xi, t)\mathcal{A}(\xi, u; \theta^*(t)) + d(t), \quad F_{\text{exec}} = f_0 + Gk_0(\xi, r, t). \quad (64)$$

Controller dynamics are included in f_0, G as needed. An allocator chooses $u \in \mathcal{U}$ using $\hat{\theta}$. Assume $\mathcal{A}(\xi, u; \hat{\theta}) - \mathcal{A}(\xi, u; \theta^*) = \Psi(\xi, u, t)(\hat{\theta} - \theta^*)$ on the operating domain. With $\delta_A = \mathcal{A}(\xi, u; \hat{\theta}) - k_0$, $B_\theta = G\Psi$ and $d_{\text{eff}} = d + G\delta_A$ give exactly (73). In a reduced-state model, $d(t)$ is a pathwise remainder that can depend on omitted physical state. A disjoint decomposition is

$$d_{\text{eff}} = d_{\text{base}} + d_{\text{hist}} + d_{\text{contact}} + G\delta_A, \\ \|d_{\text{eff}}\|_{M_c} \leq \bar{d}_{\text{base}, M} + \bar{d}_{\text{hist}, M} + \bar{d}_{\text{contact}, M} + \bar{d}_{\text{real}, M}.$$

These bounds must hold uniformly over admissible corrected states, commands, and omitted states within each certified smooth mode. Effects already represented by matching or realization error are not counted again as contact disturbance. Contact remainder bounds are assumptions, not consequences of servo stability; impacts require the separate hybrid jump conditions in Appendix B.4, not a finite continuous force bound. For gain and bias, $\mathcal{A} = \text{diag}(g)u + b$, the regressor contains command and identity columns. Estimated gains bounded away from zero and feasible commands are necessary for an inverse. Payload and friction changes need a valid parameterization or bounded remainder. A locked joint can remove authority. Perception changes can change the supplied commands; execution tracking alone does not establish that those commands remain task-correct.

The nominal reference solves $\dot{\xi}^{\text{nom}} = F_{\text{exec}}(\xi^{\text{nom}}, r(t), t)$ with a declared initial state. It is the nominal physical response to the *same commands actually supplied during deployment*, not a counterfactual rollout that recomputes VLA commands from a healthy observation stream. The reference may be an analytical comparison system for prediction-only adaptation. Computing it online is required only for a controller that uses its tracking signal. For a reduced state, this is the response of the declared nominal reduced model; a full-contact nominal simulation is a different comparison unless its projection realizes that model or its discrepancy is bounded.

For every allowed command and each certified smooth operating mode, assume F_{exec} is continuously differentiable in ξ , and $M_c(\xi, t)$ is C^2 in ξ and C^1 in t locally. Uniform positive metric bounds and (74) hold on a stated operating domain Ω uniformly over $r \in \mathcal{R}$. The derivative A_{exec} holds r fixed: it includes the robot and low-level feedback sensitivity, with no VLA derivative. Although deployment commands can arise from feedback, the nominal comparison and all virtual trajectories in this proof use the same supplied $r(t)$. This is an input-conditioned execution certificate, not a contraction certificate for the complete VLA loop. Admissibility of the actual command signal and domain containment are separate assumptions; the certificate does not establish them. Shared commands do not imply shared object states or contact modes. A hybrid comparison must cover the realized mode sequences and event timing; a smooth servo metric alone supplies only its stated servo-domain guarantee.

Take a neighborhood on which every relevant pair has a minimizing constant-speed geodesic contained in Ω , and the squared metric distance is smooth. A strongly geodesically convex normal neighborhood is a sufficient local setting. The flow perturbations in first variation must remain in Ω for sufficiently small time increments. Bounds hold only until these conditions cease. Nonunique minimizers require a separate upper-Dini argument; we do not assume a differentiable geodesic choice through a cut locus.

The metric is independent of the command, the estimated parameter, and the unknown parameter. Dependence on these quantities introduces extra derivative or switching terms and requires a different certificate. Similarly, $(\xi - \xi^{\text{nom}})^\top M_c(\xi, t)(\xi - \xi^{\text{nom}})/2$ is generally not the geodesic half-energy. With a command-independent metric, a jump of r does not itself jump the energy if the physical/reference states are continuous; the flow bound holds on each command interval and extends

across the jump by continuity. Direct reference resets, command-dependent tracking-coordinate resets, and state or metric resets need explicit jump bounds or a feasible interpolated reference. Physical controller memory, actuator delays, and contact resets must be covered by the state partition and sampled/hybrid conditions in Appendix B.4. VLA internal policy memory is not part of the primary execution certificate.

C.2 IDENTIFICATION INFORMATION AND THE AVAILABLE FEEDBACK SIGNAL

For measured or filtered regressions, declare weights $R_j \succ 0$, $\omega_j \geq 0$, and form

$$\begin{aligned} y_j &= \Phi_j \theta^*(t_j) + \varepsilon_j, & H_I &= \sum_j \omega_j \Phi_j^\top R_j^{-1} \Phi_j, \\ b_I &= \sum_j \omega_j \Phi_j^\top R_j^{-1} y_j, & \epsilon_I &= b_I - H_I \theta^*(t). \end{aligned} \quad (65)$$

Then $b_I - H_I \hat{\theta} = -H_I \tilde{\theta} + \epsilon_I$ is the negative gradient of the weighted prediction loss. Current data or a retained informative dataset can supply this statistic. A command-history FIR buffer supplies predictor context; it is not, by itself, an informative identification memory. Estimation requires observability of the physical parameter directions and a bias reference separating nominal mismatch from the change being identified.

The geodesic signal $\sigma_{\text{exec}} = B_\theta^\top p_\xi$ additionally requires the nominal physical reference, realization sensitivity, and metric endpoint covector. Appendix E.4 constructs these quantities from robot/controller dynamics. A low-level reference generated from the current VLA chunk can be used when it realizes the same nominal execution dynamics, or its dynamics defect is retained. Neither the metric nor this tracking signal requires differentiating the VLA. The evaluated prediction-only method computes neither quantity. For a computed $\hat{\sigma}_{\text{exec}} = \sigma_{\text{exec}} + \epsilon_\sigma$ with $\|\epsilon_\sigma\| \leq \bar{\epsilon}_\sigma$, the parameter forcing in (79) becomes $k_c \sqrt{\gamma_+} \bar{\epsilon}_I + \kappa \sqrt{\gamma_+} \bar{\epsilon}_\sigma + \nu / \sqrt{\gamma_-}$. Approximate metrics or geodesics must justify this error bound and any metric-inequality defect. If an approximate reference is used only to compute $\hat{\sigma}_{\text{exec}}$, its error is relative to the exact ξ^{nom} in the theorem. Tracking a different reference satisfying $\xi^{\text{nom}} = F_{\text{exec}}(\xi^{\text{nom}}, r, t) + \delta_r$ adds the separate first-variation term $-p_r^\top \delta_r$, where $p_r = M_c(\xi^{\text{nom}}, t) c_s(0)$. Its magnitude is at most $X \|\delta_r\|_{M_c(\xi^{\text{nom}}, t)}$; this reference-dynamics defect must be added to the tracking forcing $\vec{d}_M$. Approximate quantities cannot simply be substituted as exact ones.

C.3 PREDICTION-ONLY CONTRACTION AND FEEDBACK COUPLING

With $\kappa = 0$, (77) is prediction-gradient adaptation in the parameter metric Γ^{-1} . Under a common exogenous stream (H_I, b_I) , two unprojected observers have variation

$$\delta \dot{\hat{\theta}} = -k_c \Gamma H_I \delta \hat{\theta}, \quad \frac{d}{dt} \frac{1}{2} \|\delta \hat{\theta}\|_{\Gamma^{-1}}^2 = -k_c \delta \hat{\theta}^\top H_I \delta \hat{\theta}. \quad (66)$$

If $H_I \succeq \mu I$, this is conditional observer contraction at rate $k_c \mu \gamma_-$. The projected case follows from normal-cone monotonicity (Appendix E.2), without differentiating through active faces. On the robot, correction changes the state and therefore its data stream; this conditional observer statement does not certify the feedback interconnection.

To account for that dependence, suppose throughout the operating domain

$$\begin{aligned} \|M_c^{1/2} B_\theta \Gamma^{1/2}\| &\leq L_B, & \|d_{\text{eff}}\|_{M_c} &\leq d_0 + \ell_x X + \ell_z Z, \\ \|\epsilon_I\| &\leq \epsilon_0 + m_x X + m_z Z, & \|\dot{\theta}^*\| &\leq \nu. \end{aligned} \quad (67)$$

All constants are nonnegative. For this execution/estimation comparison (80), define

$$\begin{aligned} a &= \lambda - \ell_x, & b &= L_B + \ell_z, \\ c &= k_c \mu \gamma_- - k_c \sqrt{\gamma_+} m_z, & d &= k_c \sqrt{\gamma_+} m_x, & q_z &= k_c \sqrt{\gamma_+} \epsilon_0 + \nu / \sqrt{\gamma_-}. \end{aligned} \quad (68)$$

Proposition 9 (Prediction-only interconnection). *Assume the geometric, matching, projection, and information conditions above on an interval beginning at t_e , with $H_I \succeq \mu I$. The $\kappa = 0$ law obeys*

(80). Write $\mathcal{A}_c = \begin{bmatrix} -a & b \\ d & -c \end{bmatrix}$ and $r_0 = (d_0, q_z)^\top$. Componentwise,

$$\begin{bmatrix} X(t) \\ Z(t) \end{bmatrix} \leq e^{\mathcal{A}_c(t-t_e)} \begin{bmatrix} X(t_e) \\ Z(t_e) \end{bmatrix} + \int_{t_e}^t e^{\mathcal{A}_c(t-s)} r_0 ds. \quad (69)$$

If $a, c > 0$ and $ac > bd$, this comparison matrix is Hurwitz, and

$$\limsup_{t \rightarrow \infty} X(t) \leq \frac{cd_0 + bq_z}{ac - bd}, \quad \limsup_{t \rightarrow \infty} Z(t) \leq \frac{dd_0 + aq_z}{ac - bd}, \quad (70)$$

provided all assumptions remain valid for that infinite interval.

These are asymptotic comparison bounds, not startup bounds. A sufficient invariant rectangle contains $(X(t_e), Z(t_e))$ and satisfies $a\bar{X} \geq b\bar{Z} + d_0$, $c\bar{Z} \geq d\bar{X} + q_z$; its whole trajectory tube must remain in the domain. Alternatively propagate (69) directly. Information acquisition requires its own transient bound. If $d = 0$, both diagonal rates being positive suffice even for large b , although the transient may be large. The sampled coupled-error result in Appendix B.5 expresses the same feedback structure with the actual estimator and application schedule.

C.4 RECOVERY, REGIME CHANGES, AND TASK SCOPE

For Theorem 10, if $W(t_e) > W_{\text{tar}} > Q/\beta$, a sufficient additional recovery time is

$$T_{\text{rec}} = \beta^{-1} \log \frac{W(t_e) - Q/\beta}{W_{\text{tar}} - Q/\beta}. \quad (71)$$

Information-acquisition time must be added and its tube enforced. A bound $\max\{W(t_e), Q/\beta\} < R$ proves containment only if the entire radius- R tube about the same-command response lies in the certified domain and command admissibility persists. This is *execution recovery*: approach to the nominal physical response under the supplied commands, up to estimation, model, timing, and realization errors. If a separate task specification guarantees success throughout $\|\xi - \xi^{\text{nom}}\| < r_{\text{task}}$ around a task-valid nominal response, one may take $R \leq \sqrt{m}r_{\text{task}}$. Both path and terminal conditions must be included. A healthy VLA rerun generally supplies a different command sequence; comparing it requires an additional argument for command divergence or for the complete policy/plant interconnection.

For example, $\dot{z} = -z + r$ contracts at rate one for a common supplied $r(t)$ in the metric $M_c = 1$. Closing the feedback $r = 2z$ instead gives $\dot{z} = z$. Execution contraction therefore proves neither full-loop stability nor task success. Dropping a VLA derivative while claiming contraction of that full loop would be invalid. No such derivative or full-loop contraction assumption enters the primary execution theorem.

A parameter jump is distinct from bounded-rate drift. If the physical and reference states, metric, Γ , and estimate are continuous or unchanged,

$$W(t_j^+) \leq W(t_j^-) + \|\Delta\theta^*(t_j)\|_{\Gamma^{-1}}. \quad (72)$$

Restart the envelope with valid post-change information. A large H_I may encode obsolete parameters; Appendix E.5 bounds stale-data bias, reacquisition, and repeated changes. Separate reference, state, or metric resets require their own jump bounds. None of these tracking-storage results proves pairwise contraction of every augmented $(\xi, \hat{\theta})$ trajectory. That stronger claim needs a differential metric inequality for the actual augmented dynamics, including state-dependent regressors, normalizers, and metric derivatives.

D CONTINUOUS COMPOSITE ADAPTATION AS AN OPTIONAL EXTENSION

This appendix derives the geometric tracking term that can augment prediction feedback when the same-command nominal execution response is available online. The composite controller was not evaluated in the VLA experiments; the main results instead analyze the documented sampled laws. Appendix C specifies physical matching, reference, and domain assumptions, including the closure of the physical state.

D.1 AN EXECUTION METRIC GENERATES THE TRACKING SIGNAL

Let $r(t)$ be the commands actually supplied by the frozen VLA and ξ the robot and low-level controller state. The nominal physical response ξ^{nom} follows that *same command sequence*, without rerunning the VLA. For $\tilde{\theta} = \hat{\theta} - \theta^*$, physical OOD and allocation give

$$\dot{\xi} = F_{\text{exec}}(\xi, r, t) - B_{\theta}\tilde{\theta} + d_{\text{eff}}, \quad \xi^{\text{nom}} = F_{\text{exec}}(\xi^{\text{nom}}, r, t). \quad (73)$$

Here F_{exec} contains the robot and its execution controller, and $B_{\theta} = G\Psi$; allocation/model errors enter d_{eff} (Appendix C). On a geodesically convex operating domain, assume a smooth metric $M_c(\xi, t)$ independent of r and $\tilde{\theta}$, with $0 < \underline{m}I \preceq M_c \preceq \bar{m}I$, and uniformly over admissible commands,

$$\partial_t M_c + \partial_{F_{\text{exec}}} M_c + A_{\text{exec}}^{\top} M_c + M_c A_{\text{exec}} \preceq -2\lambda M_c, \quad A_{\text{exec}} = \partial_{\xi} F_{\text{exec}}|_{r, \text{fixed}}, \quad \lambda > 0. \quad (74)$$

No VLA Jacobian is required. Appendix E.4 constructs the metric from physical tracking dynamics and controller gains. For a constant-speed minimizing geodesic c from ξ^{nom} to ξ , set

$$V_{\xi} = \frac{1}{2} \int_0^1 c_s^{\top} M_c(c, t) c_s ds, \quad X = \sqrt{2V_{\xi}}, \quad p_{\xi} = M_c(\xi, t) c_s(1), \quad \sigma_{\text{exec}} = B_{\theta}^{\top} p_{\xi}. \quad (75)$$

First variation of this half-energy (Appendix E.1) gives

$$D^+ V_{\xi} \leq -\lambda X^2 - \sigma_{\text{exec}}^{\top} \tilde{\theta} + p_{\xi}^{\top} d_{\text{eff}}, \quad |p_{\xi}^{\top} d_{\text{eff}}| \leq X \|d_{\text{eff}}\|_{M_c}. \quad (76)$$

Thus the metric generates the adaptive tracking signal; for constant $M_c = P$, $\sigma_{\text{exec}} = B_{\theta}^{\top} P(\xi - \xi^{\text{nom}})$ (Lopez & Slotine, 2020; Tsukamoto et al., 2021a). The primary certificate excludes VLA policy memory; physical execution memory must satisfy the partition in Appendix B.4.

D.2 COMPOSITE EXTENSION AND THE EVALUATED PREDICTION BRANCH

A weighted regression supplies $b_I = H_I \theta^* + \epsilon_I$. With constant $\Gamma \succ 0$, its prediction gradient and the geodesic coupling suggest

$$\dot{\tilde{\theta}} = \text{Proj}_{\Theta}^{\Gamma} \left(\tilde{\theta}, \Gamma [\kappa \sigma_{\text{exec}} + k_c (b_I - H_I \tilde{\theta})] \right), \quad k_c > 0, \quad 0 \leq \kappa \leq 1. \quad (77)$$

The velocity projection uses the Γ^{-1} metric on a fixed convex set containing the true parameter and initial estimate. Adding $V_{\theta} = \|\tilde{\theta}\|_{\Gamma^{-1}}^2/2$ cancels $-\sigma_{\text{exec}}^{\top} \tilde{\theta}$ exactly at $\kappa = 1$.

Theorem 10 (Contraction-derived execution recovery). *For the same-command execution pair in (73), suppose the metric, matching, and projection assumptions hold after t_e without state, metric, or parameter jumps, with absolutely continuous θ^* , $H_I \succeq \mu I$, $\mu > 0$, $\|d_{\text{eff}}\|_{M_c} \leq \bar{d}_M$, $\|\epsilon_I\| \leq \bar{\epsilon}_I$, and $\|\dot{\theta}^*\| \leq \nu$. For (77) with $\kappa = 1$, put $\gamma_+ = \lambda_{\max}(\Gamma)$ and $\gamma_- = \lambda_{\min}(\Gamma)$. Then, while these domain assumptions hold,*

$$W(t) \leq e^{-\beta(t-t_e)} W(t_e) + \frac{Q}{\beta} (1 - e^{-\beta(t-t_e)}), \quad W^2 = X^2 + \|\tilde{\theta}\|_{\Gamma^{-1}}^2, \quad (78)$$

$$\beta = \min\{\lambda, k_c \mu \gamma_-\}, \quad Q^2 = \bar{d}_M^2 + (k_c \sqrt{\gamma_+} \bar{\epsilon}_I + \nu / \sqrt{\gamma_-})^2. \quad (79)$$

This gives $\|\xi - \xi^{\text{nom}}\| \leq W/\sqrt{m}$ and $\|\tilde{\theta}\| \leq \sqrt{\gamma_+} W$, with a noise/drift floor. This bounds physical execution relative to the supplied-command response; it does not certify a separately evolving healthy VLA rollout or task success. The composite extension requires an available ξ^{nom} and σ_{exec} ; it was not evaluated in the VLA runs.

The evaluated architecture uses prediction feedback only ($\kappa = 0$); ξ^{nom} may then remain an analytical comparison and no online geodesic is required. Its observation-attenuated law is a biased prediction step, not a literal discretization of the unbiased law above. Actual normalization, holds, and sampling are analyzed in Section 3.1. For the prediction-only continuous law,

$$\begin{bmatrix} D^+ X \\ D^+ Z \end{bmatrix} \leq \begin{bmatrix} -a & b \\ d & -c \end{bmatrix} \begin{bmatrix} X \\ Z \end{bmatrix} + \begin{bmatrix} d_0 \\ q_z \end{bmatrix}, \quad a, c > 0, \quad ac > bd, \quad Z = \|\tilde{\theta}\|_{\Gamma^{-1}}. \quad (80)$$

Appendix C.3 derives these coefficients from metric sensitivity, information, and predictor error. They express how nominal contraction and identification must dominate their feedback coupling. A stable cascade ($d = 0$) permits large one-way sensitivity b . A physical execution metric and its domain have since been measured for the reported robots: established on the GR1 arm, refuted on the Panda under operational-space control (Appendix G.15). The composite law is therefore evaluable on the humanoid arm and not on the Panda. Full-VLA contraction is not required.

E CONTRACTION-DERIVED ADAPTATION: PROOFS AND EXTENSIONS

The results concern the system and domain explicitly specified in Appendix C. Geodesic cancellation establishes adaptive tracking of the nominal physical response to the same supplied commands. The certificate concerns the robot and low-level execution controller; it does not certify the full VLA feedback loop or differential dynamics of the augmented adaptive system.

E.1 FIRST VARIATION OF THE GEODESIC HALF-ENERGY

Fix a time t within a command-continuity interval in a certified smooth mode, and the minimizing geodesic $c(s)$ in (75). For a small positive increment h , flow each point under F_{exec} with the same supplied VLA command signal: $c_h(s) = \phi_{t+h,t}^r(c(s))$. The signal is fixed for every virtual trajectory. The curve and its short-time images remain in that smooth domain; hybrid crossings require the separate bounds in Appendix B.4. The tangent variation obeys $\partial_h(c_h)_s = A_{\text{exec}}(c_h, r(t+h), t+h)(c_h)_s$. No policy derivative enters this variation. Write its energy as $\mathcal{E}(h) = \frac{1}{2} \int_0^1 (c_h)_s^\top M_c(c_h, t+h)(c_h)_s ds$. Differentiating at $h = 0$ gives

$$\begin{aligned} \mathcal{E}'(0) &= \frac{1}{2} \int_0^1 c_s^\top (\partial_t M_c + \partial_{F_{\text{exec}}} M_c + A_{\text{exec}}^\top M_c + M_c A_{\text{exec}}) c_s ds \\ &\leq -\lambda \int_0^1 c_s^\top M_c c_s ds = -\lambda X^2. \end{aligned} \quad (81)$$

The minimizing energy between the transported endpoints is no larger than that of this transported curve. They agree at $h = 0$, so the same upper derivative bound applies to the squared distance when both endpoints follow F_{exec} under the same supplied $r(t)$.

For completeness, with the start point fixed, perturb the endpoint by $\delta\xi$ and let $\delta c(s)$ be the induced curve variation. The first variation at a minimizing geodesic is

$$\delta V_\xi = [c_s^\top M_c \delta c]_0^1 + \int_0^1 \left[\frac{\partial L}{\partial c} - \frac{d}{ds} \frac{\partial L}{\partial c_s} \right]^\top \delta c ds = p_\xi^\top \delta \xi, \quad L = \frac{1}{2} c_s^\top M_c(c, t) c_s. \quad (82)$$

The interior term vanishes by the geodesic Euler–Lagrange equation; $\delta c(0) = 0$ and $\delta c(1) = \delta \xi$. Therefore the actual endpoint perturbation $-B_\theta \tilde{\theta} + d_{\text{eff}}$ adds exactly $-\sigma_{\text{exec}}^\top \tilde{\theta} + p_\xi^\top d_{\text{eff}}$ to the nominal upper derivative. This proves (76). The smooth squared distance assumption makes this a classical endpoint derivative; writing D^+ also accommodates norm comparisons at zero. We have not assumed smoothness through a cut locus. For piecewise-continuous commands, the flow inequality applies almost everywhere and is concatenated across command changes. If ξ , ξ^{nom} , and M_c are continuous, these changes cause no energy jump because the metric is command-independent.

Since the minimizing geodesic has constant metric speed on $[0, 1]$, $c_s(1)^\top M_c(\xi, t) c_s(1) = X^2$. Hence

$$\|M_c(\xi, t)^{-1/2} p_\xi\| = X, \quad |p_\xi^\top d_{\text{eff}}| \leq X \|d_{\text{eff}}\|_{M_c(\xi, t)}, \quad |\sigma_{\text{exec}}^\top \tilde{\theta}| \leq L_B X Z. \quad (83)$$

The last inequality uses (67) and $Z = \|\tilde{\theta}\|_{\Gamma^{-1}}$. The metric lower bound also gives $X \geq \sqrt{m} \|\xi - \xi^{\text{nom}}\|$, since the Euclidean length of any connecting curve is at least the endpoint distance.

E.2 PROJECTION, CANCELLATION, AND THE COMPOSITE BOUND

Let Θ be fixed, closed, and convex, with initial estimate and true parameter in Θ . Define $\text{Proj}_\Theta^\Gamma(h, v)$ as the metric projection of v onto the tangent cone $T_\Theta(h)$ in the Γ^{-1} norm. Assume the resulting projected differential equation admits an absolutely continuous solution between jumps. The weighted tangent-cone projection yields a normal $n = \Gamma^{-1}(v - \text{Proj}_\Theta^\Gamma(h, v)) \in N_\Theta(h)$, where $n^\top(\theta - h) \leq 0$ for all $\theta \in \Theta$. Taking $h = \hat{\theta}$ and $\theta = \theta^*$ proves

$$\tilde{\theta}^\top \Gamma^{-1} [\text{Proj}_\Theta^\Gamma(\hat{\theta}, v) - v] \leq 0. \quad (84)$$

This parameter projection does not ensure feasible actuator commands; allocation must enforce the physical input constraints separately.

With $V_\theta = \tilde{\theta}^\top \Gamma^{-1} \tilde{\theta}/2$, the update (77) and (84) imply, almost everywhere,

$$\dot{V}_\theta \leq \kappa \tilde{\theta}^\top \sigma_{\text{exec}} - k_c \tilde{\theta}^\top H_I \tilde{\theta} + k_c \tilde{\theta}^\top \epsilon_I - \tilde{\theta}^\top \Gamma^{-1} \dot{\theta}^*. \quad (85)$$

Thus $V = V_\xi + V_\theta$ obeys the derived cancellation identity

$$\begin{aligned} D^+ V \leq & -\lambda X^2 - (1 - \kappa) \sigma_{\text{exec}}^\top \tilde{\theta} - k_c \tilde{\theta}^\top H_I \tilde{\theta} + p_\xi^\top d_{\text{eff}} \\ & + k_c \tilde{\theta}^\top \epsilon_I - \tilde{\theta}^\top \Gamma^{-1} \dot{\theta}^*. \end{aligned} \quad (86)$$

At $\kappa = 1$ the cross term cancels, which determines the positive sign and endpoint form of the tracking term in the law. With $\sqrt{\gamma_-} Z \leq \|\tilde{\theta}\| \leq \sqrt{\gamma_+} Z$, $\|\Gamma^{-1/2} \dot{\theta}^*\| \leq \nu/\sqrt{\gamma_-}$, and the bounds in Theorem 10,

$$D^+ V \leq -\beta(X^2 + Z^2) + \bar{d}_M X + (k_c \sqrt{\gamma_+} \bar{\epsilon}_I + \nu/\sqrt{\gamma_-}) Z \leq -2\beta V + Q\sqrt{2V}. \quad (87)$$

For $W = \sqrt{2V} > 0$, divide by W . At zero, regularize the norm by $\sqrt{2V + \eta}$ and let $\eta \downarrow 0$ to obtain the same upper comparison. Integrating $D^+ W \leq -\beta W + Q$ proves (78). Euclidean component bounds follow from metric bounds, and tube containment is the standard first-exit argument under all stated domain conditions. This proves Theorem 10. If $\hat{\sigma}_{\text{exec}} = \sigma_{\text{exec}} + \epsilon_\sigma$ is used, add $\kappa \tilde{\theta}^\top \epsilon_\sigma$ to (86); its norm bound gives exactly the extra forcing specified in Appendix C. A metric inequality with certified defect $+2\delta M_c$ similarly replaces λ by $\lambda - \delta$ when this rate remains positive.

Projected observer contraction requires a common data stream. For two prediction-only solutions h_1, h_2 , write $\dot{h}_i = k_c \Gamma(b_I - H_I h_i) - \Gamma n_i$ with $n_i \in N_\Theta(h_i)$. Normal-cone monotonicity gives $(h_1 - h_2)^\top (n_1 - n_2) \geq 0$. Therefore

$$\frac{d}{dt} \frac{1}{2} \|h_1 - h_2\|_{\Gamma^{-1}}^2 \leq -k_c (h_1 - h_2)^\top H_I (h_1 - h_2) \leq -k_c \mu \gamma_- \|h_1 - h_2\|_{\Gamma^{-1}}^2. \quad (88)$$

This proof applies to the projected differential equation without classical Jacobians at active faces. It compares observers driven by identical exogenous H_I, b_I ; it does not discard their state dependence in the actual robot loop.

E.3 PREDICTION-ONLY COMPARISON AND THE WEIGHTED EXTENSION

For $\kappa = 0$, divide (76) by $X > 0$ and (85) by $Z > 0$; apply (83) and (67). This yields

$$\begin{aligned} D^+ X & \leq -(\lambda - \ell_x)X + (L_B + \ell_z)Z + d_0, \\ D^+ Z & \leq k_c \sqrt{\gamma_+} m_x X - (k_c \mu \gamma_- - k_c \sqrt{\gamma_+} m_z)Z + q_z. \end{aligned} \quad (89)$$

At zero distance, the same inequalities follow from the upper directional derivative of the metric norm: the coincident endpoints separate at most at the metric norm of their velocity difference. The parameter norm is handled in the same way, or by regularization. The off-diagonal entries are nonnegative, so the comparison matrix is Metzler. Positivity of its matrix exponential and variation of constants prove (69). A two-by-two Metzler matrix with positive a, c is Hurwitz exactly when $ac > bd$. Its inverse gives the corner $-\mathcal{A}_c^{-1} r_0$ in (70). On a rectangle's upper faces the two derivative inequalities are nonpositive under the stated startup conditions, proving its invariance. This establishes Proposition 9.

For intermediate κ and coupled model error, one weighted storage retains the cancellation mechanism without adding a competing theorem.

Proposition 11 (Weighted contraction-derived tracking certificate). *Assume the geometric, matching, projection, and information conditions of Theorem 10 on a no-jump interval, replacing its exogenous error bounds by (67). Take $\tau > 0$ and define $W_\tau^2 = \tau X^2 + Z^2$ and*

$$h_{\kappa, \tau} = \frac{|\kappa - \tau| L_B + \tau \ell_z + k_c \sqrt{\gamma_+} m_x}{\sqrt{\tau}}, \quad K_{\kappa, \tau} = \begin{bmatrix} a & -h_{\kappa, \tau}/2 \\ -h_{\kappa, \tau}/2 & c \end{bmatrix}, \quad (90)$$

$$\beta_{\kappa, \tau} = \frac{1}{2} \left[a + c - \sqrt{(a - c)^2 + h_{\kappa, \tau}^2} \right], \quad Q_\tau^2 = \tau d_0^2 + q_z^2. \quad (91)$$

Here a, c, q_z are as in (68). If $K_{\kappa, \tau} \succ 0$, the exponential envelope (78) holds for W_τ with rate $\beta_{\kappa, \tau}$ and forcing Q_τ . The component bounds are $\|\xi - \xi^{\text{nom}}\| \leq W_\tau/\sqrt{\tau m}$ and $\|\tilde{\theta}\| \leq \sqrt{\gamma_+} W_\tau$.

Proof. For $V_\tau = \tau V_\xi + V_\theta$, the uncanceled cross term is $(\kappa - \tau)\sigma_{\text{exec}}^\top \tilde{\theta}$. Its magnitude is at most $|\kappa - \tau|L_B XZ$. Substituting the coupled-error bounds and $X_\tau = \sqrt{\tau}X$ gives $D^+V_\tau \leq -aX_\tau^2 - cZ^2 + h_{\kappa,\tau}X_\tau Z + \sqrt{\tau}d_0X_\tau + q_zZ$. The eigenvalue and norm argument in (87) proves the claim. $\square$

With approximate tracking signal, replace q_z by $q_z + \kappa\sqrt{\gamma_+}\bar{e}_\sigma$. For $\kappa = 0$, $h_{0,\tau} = b\sqrt{\tau} + d/\sqrt{\tau}$. If $b, d > 0$, its minimum is $2\sqrt{bd}$ at $\tau = d/b$; positive definiteness is therefore feasible exactly when $ac > bd$ with $a, c > 0$. If either off-diagonal coefficient vanishes, a sufficiently small or large positive weight gives the stable-cascade result. Weights affect certificate conservatism and the transient bound, not the controller. A failed unweighted test does not imply instability. At $\kappa = 1$, $\tau = 1$, and zero endogenous couplings this proposition reduces to the composite theorem in Appendix D. This rate comparison is conditional on common valid bounds; changed controllers may change the actual regressors and errors.

E.4 OBTAINING THE METRIC FROM THE ROBOT AND EXECUTION CONTROLLER

The principal route is to establish a contraction metric for the existing robot/controller tracking dynamics. The uncontrolled plant need not contract: $\dot{q} = u$ preserves the separation between trajectories under a common input and has no strict contraction with a uniformly bounded positive metric on an infinite time interval. Execution feedback supplies the required decay.

For linear physical execution-error coordinates $e = \xi - \xi^{\text{nom}}$, suppose

$$\dot{e} = A_{\text{cl}}(t)e - B_\theta(t)\tilde{\theta} + d_{\text{eff}}, \quad \dot{P} + A_{\text{cl}}^\top P + PA_{\text{cl}} \preceq -2\lambda P, \quad P(t) \succ 0. \quad (92)$$

With uniform positive bounds on P , $M_c = P(t)$ is an execution contraction metric. The geodesic is straight, so $V_\xi = e^\top Pe/2$ and $\sigma_{\text{exec}} = B_\theta^\top Pe$. For example, nominal PD tracking of a fully actuated double integrator has the constant difference dynamics

$$A_{\text{cl}} = \begin{bmatrix} 0 & I \\ -K_p & -K_d \end{bmatrix}, \quad A_{\text{cl}}^\top P + PA_{\text{cl}} = -Q_{\text{PD}}, \quad \lambda = \frac{1}{2}\lambda_{\min}(P^{-1/2}Q_{\text{PD}}P^{-1/2}). \quad (93)$$

If A_{cl} is Hurwitz, choosing $Q_{\text{PD}} \succ 0$ gives a unique $P \succ 0$. Here Q_{PD} is a design matrix, distinct from the scalar forcing bound Q in Theorem 10. The supplied command can enter the linear nominal dynamics as a common forcing and cancels from their difference. The calculation uses robot dynamics and controller gains; it never differentiates the VLA.

For a nonlinear robot, a stable linearization is insufficient. One must establish (74) for the actual execution vector field throughout the operating region and uniformly over admissible commands, with remainders included where appropriate. A control contraction metric may instead support designing a suitable feedback controller (Manchester & Slotine, 2017); this is an additional controller design, not a certificate already established for the existing controller. The sampled servo in Appendix F provides one verified discrete execution instance. Neither construction certifies the contact-rich VLA robot/controller without checking its dynamics and physical state closure (Appendix B.4). In particular, a PD servo certificate remains a certificate of the modeled servo if object/contact effects are omitted. Extending its tube through manipulation requires a uniform bound on those effects or a metric for the enlarged physical state; impacts and differing event times need the corresponding hybrid transition bounds.

An explicit first-order execution channel. For an available differentiable target q_d generated from the supplied command, consider

$$\dot{q} = A(\theta^*)u + b(\theta^*) + d, \quad k_0(q, r, t) = \dot{q}_d - K(q - q_d), \quad K = K^\top \succ 0. \quad (94)$$

Let the nominal response satisfy $\dot{\xi}^{\text{nom}} = \dot{q}_d - K(\xi^{\text{nom}} - q_d)$, with $\xi = q$. If the coefficient model is affine in θ and the allocator realizes $A(\hat{\theta})u + b(\hat{\theta}) = k_0 + \delta_A$ with feasible u , then

$$\dot{e} = -Ke - \Psi(u)\tilde{\theta} + d + \delta_A, \quad M_c = I, \quad \sigma_{\text{exec}} = \Psi(u)^\top e, \quad \lambda = \lambda_{\min}(K), \quad e = q - \xi^{\text{nom}}. \quad (95)$$

Initializing $\xi^{\text{nom}}(t_0) = q_d(t_0)$ makes the nominal response equal to q_d for this exact channel; otherwise it retains its nominal transient. This equality must not be assumed for an arbitrary

commanded target. A discontinuous q_d must be replaced by a feasible differentiable reference, or the corresponding reset and dynamics defect must be bounded. Prediction-only adaptation need not integrate the nominal response or compute e online.

For unrestricted matrix and offset coefficients, $\Psi(u) = [u^\top \otimes I, I]$; a structured model retains appropriate columns. Finite command bounds give finite L_B . Integrated measured motion supplies a regression, for example $q(t) - q(t-h) = \int_{t-h}^t \Psi(u(s)) ds \theta^*(t) + \varepsilon_t$, where ε_t includes disturbances, sensing error, and parameter variation within the window. For an affine model with a known nominal offset, subtract its integrated contribution first. This avoids assuming noiseless derivatives. Tracking a filtered VLA target in this construction remains a physical execution guarantee, without a task-success claim.

E.5 INFORMATION, CHANGING REGIMES, AND REACQUISITION

In a constant regime, retained samples can maintain $H_I \succeq \mu I$ after new commands cease to excite the system, provided the data stay valid and their weights do not decay away. This is the finite-excitation memory mechanism (Chowdhary & Johnson, 2010; Parikh et al., 2019); it creates no missing information. Because $R_j \succ 0$, $H_I v = 0$ implies $\Phi_j v = 0$ at each positively weighted sample. Those data cannot distinguish θ from $\theta + v$. Tracking and full parameter convergence therefore have different information requirements.

Proposition 12 (Data age and parameter jumps). *For (65), the exact identity is*

$$\epsilon_I(t) = \sum_j \omega_j \Phi_j^\top R_j^{-1} \varepsilon_j + \sum_j \omega_j \Phi_j^\top R_j^{-1} \Phi_j [\theta^*(t_j) - \theta^*(t)]. \quad (96)$$

For past samples $t_j \leq t$, within-regime drift at most ν , and parameter jumps $\Delta\theta_\ell$ at t_ℓ , consequently

$$\begin{aligned} \|\epsilon_I(t)\| &\leq \sum_j \omega_j \|\Phi_j^\top R_j^{-1}\| \|\varepsilon_j\| \\ &\quad + \sum_j \omega_j \|\Phi_j^\top R_j^{-1} \Phi_j\| \left[\nu(t - t_j) + \sum_{\ell: t_j < t_\ell \leq t} \|\Delta\theta_\ell\| \right]. \end{aligned} \quad (97)$$

If a jump leaves the physical/reference state, metric, estimate, and Γ continuous or unchanged, (72) holds for W and for W_τ .

Proof. Substitute the measured regression into $b_I - H_I \theta^*(t)$, then bound parameter displacement by integrated drift and intervening jumps. At a jump, $\tilde{\theta}^+ = \tilde{\theta}^- - \Delta\theta$; apply the triangle inequality to $(X, \Gamma^{-1/2} \tilde{\theta})$ or $(\sqrt{\tau} X, \Gamma^{-1/2} \tilde{\theta})$. $\square$

Clearing data causes no direct storage jump, but can destroy $\mu > 0$. Bounded-age memory and forgetting limit stale bias only with sufficient new information to maintain a rate. Reset must use available signals, with a specified detection delay and false-reset policy; no perfect fault-onset detector is assumed.

Before information is acquired, the weighted dissipation remains valid with a time-dependent, possibly negative lower rate $\underline{\beta}(t) = \lambda_{\min} K_{\kappa, \tau}(t)$ and forcing $Q_\tau(t)$. For $\tau = 1$ to simplify notation,

$$W(t) \leq e^{-\int_{t_0}^t \underline{\beta}(s) ds} W(t_0) + \int_{t_0}^t e^{-\int_s^t \underline{\beta}(v) dv} Q_1(s) ds. \quad (98)$$

If $\underline{\beta} \geq -g$, $g \geq 0$, and $Q_1 \leq Q_a$ for an acquisition interval of length at most T_e , then

$$W_{\text{after}} \leq A_e W_{\text{before}} + D_e, \quad A_e = e^{gT_e}, \quad D_e = \begin{cases} Q_a(e^{gT_e} - 1)/g, & g > 0, \\ Q_a T_e, & g = 0. \end{cases} \quad (99)$$

With exact composite cancellation and bounded exogenous errors, $g = 0$ is valid for merely positive-semidefinite H_I . Without cancellation or with endogenous error this conclusion is not automatic. In every case the acquisition envelope must remain inside the operating domain.

For repeated changes, let $B_n = W(t_n^-)$, jumps be at most J , acquisition obey (99), and the following informative interval have length at least T_s , rate $\beta > 0$, and floor $W_\infty = Q/\beta$. For $r = e^{-\beta T_s}$ and $C_n = [B_n - W_\infty]_+$, stagewise propagation yields

$$C_{n+1} \leq rA_e C_n + r[(A_e - 1)W_\infty + A_e J + D_e]. \quad (100)$$

The acquisition excess is at most the nonnegative bracketed expression plus $A_e C_n$, and informative decay multiplies its positive part by at most r . If $rA_e < 1$, the limiting pre-jump excess is bounded by the second term divided by $1 - rA_e$. Faster regime switching can defeat recovery even when each informative regime has a positive rate.

E.6 IDENTIFIABLE EFFECTS, CORRECTION AUTHORITY, AND DECODER REALIZATION

For $a_{\text{eff}} = A(\theta^*, \xi)u + b(\theta^*, \xi)$, exact realization of a desired interface command a requires $a - b(\theta^*, \xi) \in A(\theta^*, \xi)\mathcal{U}$. For the continuous execution model, this a equals the low-level demand $k_0(\xi, r, t)$, rather than a separately evaluated VLA policy. At a post-decoder action interface it instead denotes the action to be realized. The allocator’s predicted residual $\delta_A = A(\hat{\theta}, \xi)u + b(\hat{\theta}, \xi) - a$ gives the exact identity

$$a_{\text{eff}} - a = [A(\theta^*, \xi) - A(\hat{\theta}, \xi)]u + b(\theta^*, \xi) - b(\hat{\theta}, \xi) + \delta_A. \quad (101)$$

For affine coefficients the first two terms supply the matched regressor. For a diagonal gain and offset, an unsaturated inverse $u_i = (a_i - \hat{b}_i)/\hat{g}_i$ gives $a_{\text{eff},i} - a_i = (g_i - \hat{g}_i)u_i + (b_i - \hat{b}_i)$. The inverse requires $\hat{g}_i$ bounded away from zero and feasible u_i . Near-zero true effectiveness may require an infeasible command; a locked direction needs surviving actuator redundancy or a changed task/reference.

Proposition 13 (Information and local correction are different conditions). *Let local physical changes have action effect $B_f\theta$, measured calibration effect $C_f\theta$, and available correction Jv . In this fixed linearization, identifying the required physical effect from the measurement for every θ is possible only if $\ker C_f \subseteq \ker B_f$. Unconstrained cancellation of every physical effect requires range $B_f \subseteq \text{range } J$. For one effect v_f , the minimum Euclidean action residual is $\|(I - JJ^\dagger)v_f\|$. Therefore a q -dimensional correction vector must satisfy $q \geq \text{rank } B_f$ to cover the whole family; this dimension condition alone is not sufficient.*

Proof. If $w \in \ker C_f$ but $B_fw \neq 0$, then θ and $\theta + w$ produce identical measurements but require different corrections. Conversely, kernel inclusion defines a well-defined linear map $L : C_f\theta \mapsto B_f\theta$ on range C_f , extendable to the whole measurement space, so the condition is also sufficient for unconstrained noise-free effect reconstruction in this linearization. Exact cancellation requires solving $Jv = -B_f\theta$; the range condition is equivalent to solvability for every θ . Orthogonal projection gives the minimum residual. Finally $\text{rank } J \leq q$ gives the dimension bound. $\square$

Constraints on v or u , noise sensitivity, changing Jacobians, and parameter identifiability remain additional requirements. In particular, full parameter recovery from $H_I \succeq \mu I$ is stronger than recovery of the action effect in this proposition. Redundant parameterizations should be reduced to identifiable physical directions before applying the full parameter theorem. The bound concerns action directions within the chosen model, not a universal minimum number of neural-network weights.

Native neural bias edits need an action-level realization bound. Let $\mathcal{D}(o, b, \omega_{\text{dec}})$ be the final decoded action for a fixed observation o and decoder random draw ω_{dec} . Edit a native bias as $b = b_0 - P_b\hat{f}$, while the intended external correction is $-D\hat{f}$. With $J_b = \partial_b \mathcal{D}(o, b_0, \omega_{\text{dec}})$, define

$$\zeta = \mathcal{D}(o, b_0 - P_b\hat{f}, \omega_{\text{dec}}) - \mathcal{D}(o, b_0, \omega_{\text{dec}}) + D\hat{f}. \quad (102)$$

If the bias Jacobian is L_b -Lipschitz along the entire bias segment in the same action units, Taylor’s integral remainder gives

$$\|\zeta\| \leq \|D - J_b P_b\| \|\hat{f}\| + \frac{L_b}{2} \|P_b \hat{f}\|^2. \quad (103)$$

The first term is linear mismatch between native coordinates and desired action directions; the second is decoder nonlinearity. The proof expands $\mathcal{D}$ along the segment and integrates $\|J_b(b_0 - sP_b\hat{f}) -$

$J_b(b_0) \leq sL_b \|P_b \hat{f}\|$. Uniform bounds require a stated observation/randomness domain. The same observation and draw isolate adapter realization from trajectory changes. Action normalization must be included in $\mathcal{D}$; nonsmooth clipping needs a direct residual bound or a separate allocation term. After the plant input map, this error contributes to d_{eff} . It must be measured or bounded before an external-addition guarantee can be assigned to native decoder editing. A registered measurement instantiates the bound for the evaluated decoder: at a fixed observation the edit is deterministic and linear (relative linear mismatch .035, quadratic part below 8%), but the gain varies 27–41% across observations, so one mean-Jacobian bias vector realizes an intended correction with median relative error 43%.

E.7 REPRODUCIBLE CONSTRUCTED CONTROL EXAMPLE

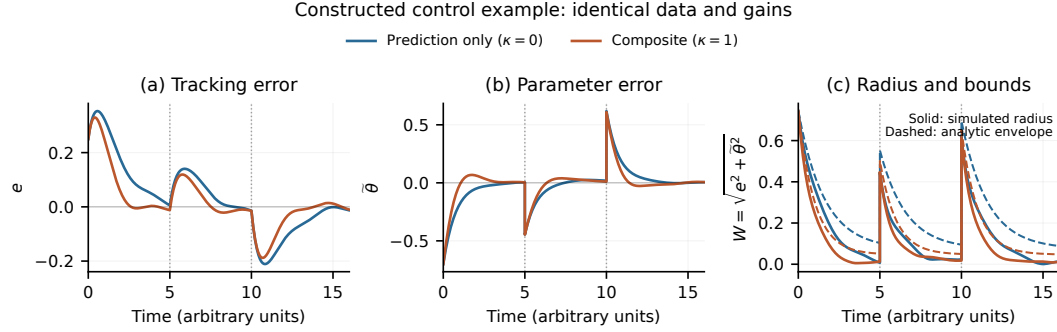


Figure 3: Constructed scalar control example. Prediction-only and composite updates use identical gains and ideal current regression data. Dashed curves are the analytic joint-error envelopes, with parameter-jump terms at times 5 and 10. This example isolates the tracking-feedback mechanism and is separate from the reported VLA experiments.

The constructed example has physical realization $\dot{q} = u + \theta^* + d$, supplied target $r = 0$, low-level feedback $k_0 = -q$, and feasible unrestricted command $u = -q - \hat{\theta}$. The same-command nominal response satisfies $\xi^{\text{nom}} = -\xi^{\text{nom}}$ with $\xi^{\text{nom}}(0) = 0$, hence $\xi^{\text{nom}} = 0$. Thus $e = q$, $V_\xi = e^2/2$, and $\sigma_{\text{exec}} = e$. Its scalar error dynamics and update are

$$\begin{aligned} \dot{e} &= -e - (\hat{\theta} - \theta^*) + .02 \sin(2.3t), \\ \dot{\hat{\theta}} &= \kappa e + 2(b_I - .65\hat{\theta}), \quad b_I = .65\theta^* + .006 \sin(1.7t + .4). \end{aligned} \quad (104)$$

with $e(0) = .25$, $\hat{\theta}(0) = 0$, and

$$\theta^*(t) = \begin{cases} .7, & 0 \leq t < 5, \\ 1.15 - .03(t - 5), & 5 \leq t < 10, \\ .4, & 10 \leq t \leq 16. \end{cases} \quad (105)$$

The statistic b_I is generated as an ideal current regression measurement, with $H_I = .65$ throughout. The adaptive update receives that statistic, not the unknown parameter directly. Both branches use identical information, disturbances, and gains. This assumption isolates the adaptive feedback mechanism; it makes no assertion about availability of comparable information on a VLA task. Projection is inactive in this example.

The jump sizes are $+.45$ at $t = 5$ and $-.6$ at $t = 10$. Between jumps, $|\dot{\theta}^*| \leq .03$, so $Q = \sqrt{.02^2 + (2 \cdot .006 + .03)^2} = .0465188$. The constant contraction metric is $M_c = 1$. The weighted-certificate matrices are $K_{0,1} = \begin{bmatrix} 1 & .5 \\ -.5 & 1.3 \end{bmatrix}$ and $K_{1,1} = \text{diag}(1, 1.3)$. We propagate the weighted envelope in Proposition 11 with $\tau = 1$ on each interval and add the jump norm at its boundary. The common drift bound also covers the constant-parameter intervals and is conservative there.

Quantity	Prediction only	Composite
Sufficient rate $\beta_{\kappa,1}$	.627985	1.000000
Radius floor $Q/\beta_{\kappa,1}$	.074076	.046519
$\int_0^{16} e(t)^2 dt$	.292041	.171081
$\int_0^{16} \tilde{\theta}(t)^2 dt$	.402695	.298536
Peak $ e $	.353284	.329610

The trajectories use DOP853 with relative tolerance 10^{-11} , absolute tolerance 10^{-13} , and maximum integration step .025. Both satisfy the analytic envelopes on the evaluated time grid. The composite parameter transient overshoots; a better joint tracking certificate is not a pointwise ordering of parameter errors. The script `audit/verify_adaptive_ood.py` regenerates the figure, trajectory arrays, constants, and numerical checks without external data.

A second exact example illustrates conservatism. For $\dot{e} = -.2e - \tilde{\theta}$, $\dot{\tilde{\theta}} = -.1\tilde{\theta}$, the prediction-only system has eigenvalues $-.2, -.1$, but the unweighted certificate has minimum eigenvalue $-.35249$. Choosing $\tau = .01$ in Proposition 11 yields minimum eigenvalue $.07929 > 0$. Thus a failed unweighted test cannot establish instability or a need for control-error feedback. A separate stale-data check uses $H_I = 2$ with old parameter .4 and current parameter 1.1, giving $\epsilon_I = -1.4$; prediction adaptation to this old target converges to .4 despite positive information. It is a data-validity diagnostic, not an evaluated reset scheme.

The script additionally checks 5,000 constructed inequalities for storage, projection, and parameter jumps with heterogeneous gains. It also checks 500 stale-memory identities. Finite numerical checks support consistency but do not replace the proofs or validate their assumptions on the VLA experiments.

E.8 RELATION TO THE EVALUATED DISCRETE OBSERVERS

For calibrated additive observations $z_k = f_k + w_k$, take $\Phi_k = I$ and $\theta^* = f$. Identification of offsets need not require persistent command motion, conditional on a separately valid healthy bias reference and identifiable residual sensitivity S . The innovation law is a sampled prediction-gradient update with $H_{I,k} = I$, but its state-dependent gain must be treated in the actual sampled proof. Its error-to-truth recursion does not establish global pairwise Euclidean contraction. The legacy observation-attenuated law targets $s_k z_k$ and has effective prediction bias $(s_k - 1)f_k + s_k w_k$. Under a fixed noiseless input, it can contract to an incorrect physical target. A gain estimator instead needs informative command regressors. None of the documented VLA updates computes the $\kappa\sigma_{\text{exec}}$ term.

The actual-law proofs in Appendix A and Appendix B retain gates, projection, history, mask, and applied snapshot age. Their coupled comparison is the sampled counterpart of Proposition 9; this connection does not replace checking those implementation conditions.

For completeness, a continuous-time bound supports an Euler claim only under additional structure. For a constant true parameter and a constant quadratic composite metric let $z = (P^{1/2}e, \Gamma^{-1/2}\tilde{\theta})$, and suppose the implemented joint map is exactly

$$z_{k+1} = \Pi_{\mathcal{Z}}[z_k + h\mathcal{F}_k(z_k) + h\delta_k], \quad 0 \in \mathcal{Z}, \quad \|\delta_k\| \leq \bar{\delta}, \quad (106)$$

where $\mathcal{Z} = \mathbb{R}^{\dim e} \times \Gamma^{-1/2}(\Theta - \theta^*)$ is closed and convex. If $z^\top \mathcal{F}_k(z) \leq -\beta\|z\|^2$ and $\|\mathcal{F}_k(z)\| \leq L\|z\|$ on the relevant domain, then

$$\|z_k\| \leq q_h^k \|z_0\| + \frac{h\bar{\delta}}{1 - q_h} (1 - q_h^k), \quad q_h = \sqrt{1 - 2\beta h + L^2 h^2} < 1, \quad 0 < h < 2\beta/L^2. \quad (107)$$

Projection is nonexpansive and fixes zero; expanding $\|z + h\mathcal{F}_k(z)\|^2$ gives the factor and summing the scalar recursion proves the claim. A physical sampled plant or chunked policy usually has a different map and needs an explicit discretization defect or a direct discrete storage inequality. Increasing gains can increase L as well as β , tightening the step-size restriction. This illustration does not certify the experimental control rate.

F A CONTRACTION-CERTIFIED EXECUTION-TRACKING CONSTRUCTION

This constructed servo demonstrates how the sampled conditions produce a useful, independently reproducible recovery envelope. It is not an additional VLA evaluation. The certificate concerns execution tracking under a common command reference, not contraction of the augmented adaptive system or a contact-rich policy.

Plant and exact metric. At 20 ms command intervals, let $\xi_k \in \mathbb{R}^2$ be position in meters, $c_k = (.06 \sin(.6t_k), .03 \cos(.4t_k))^\top$ a supplied position reference, and $t_k = .02k$. Define

$$\begin{aligned}\xi_{k+1} &= A\xi_k + u_k + f_k, & \xi_{k+1}^{\text{nom}} &= A\xi_k^{\text{nom}} + a_k, \\ a_k &= (I - A)c_k, & u_k &= \text{clip}_{[-.025, .025]^2}(a_k - D\hat{f}_{\tau_k}).\end{aligned}\tag{108}$$

$$P = \text{diag}(1, 4), \quad A = .92P^{-1/2} \text{Rot}(.04)P^{1/2}.\tag{109}$$

Here A represents the nominal robot with its execution controller, not an uncontrolled integrator. The supplied command in Theorem 2 is a_k ; the nominal physical copy receives this same sequence. The position target c_k is not itself the nominal response ξ_k^{nom} . Commands and faults are additive position increments in meters. Orthogonality of Rot gives $A^\top P A = .92^2 P$ globally. Thus $M_{c,k} = P$, $X_k = \|\xi_k - \xi_k^{\text{nom}}\|_P$, $\lambda_k^{\text{exec}} = .92$, and $L_k = \sqrt{\lambda_{\max}(P)} = 2$ satisfy the sufficient smooth-mode condition in Eq. (45), hence Eq. (13), and the input bound for every state. These constants are analytical; no trajectory fit or sampled Jacobian search is used. The equivalent nominal forgetting rate is $-\log(.92)/.02 = 4.169 \text{ s}^{-1}$.

Causal proprioception and imperfect centering. After command k , the measured execution increment is $v_k = \xi_{k+1} - A\xi_k = u_k + f_k$. A seven-tap proprioceptive filter and its healthy predictor produce

$$\begin{aligned}y_k &= S \sum_{\ell=0}^6 h_\ell v_{k-\ell} + Sb, & \hat{y}_k &= S \sum_{\ell=0}^6 h_\ell u_{k-\ell}, \\ z_k &= S^{-1}(y_k - \hat{y}_k) = \sum_{\ell=0}^6 h_\ell f_{k-\ell} + b.\end{aligned}\tag{110}$$

where $S = \text{diag}(1, .7)$, $h = (.40, .25, .15, .10, .05, .03, .02)$, and $b = (.0003, -.0002)^\top \text{ m}$ is unre-moved sensor-centering bias. The fault is zero before $k = 50$ and $f_k = (.006, .0025)^\top \text{ m}$ thereafter. Prehistory has $u_k = v_k = f_k = 0$ for $k < 0$. Therefore the onset contribution to $w_k = z_k - f_k$ is exactly $-\sum_{\ell=j+1}^6 h_\ell f_{50}$ at $k = 50 + j$, $0 \leq j \leq 6$; it vanishes in response $k = 56$, acquired at 1.14 s, which is 120 ms after the first post-fault response. Observations are plotted at their acquisition times t_{k+1} . The estimator uses measured z_k , never injected fault truth.

Actual update, projection, and application. We apply Eqs. (3) and (4) with $P_r = I_2$, $\gamma = .08$, $\rho = .015 \text{ m}$, $\delta = .0008 \text{ m}$, and coordinatewise projection onto $\mathcal{C} = [-.015, .015]^2 \text{ m}$. The innovation gain depends on the current estimate exactly as in Eq. (4). Both coordinates are estimated in every adaptive condition. Full correction uses $D = I_2$; the insufficient-support condition uses $D = \text{diag}(1, 0)$ and the same innovation estimator. The frozen condition has $D = 0$ and no updates. All states and estimates start at zero. Across the 600 commands, $\chi_k = 0$ for $65 \leq k < 100$ and whenever $k \bmod 10 = 9$, and is one otherwise. These are preset clock-based missed updates, independent of fault truth. A five-command application chunk uses $\tau_k = \max\{0, 5\lfloor k/5 \rfloor - 2\}$; snapshot age is at most six steps. An estimate updated after response k first exists at index $k + 1$. Projection and command clipping are implemented separately. Convexity keeps the update targets and estimates inside $\mathcal{C}$; furthermore $|I - A|(.06, .03)^\top + .0151 < (.02206, .01854)^\top < .0251$. Neither projection nor command clipping activates, even if the construction is continued indefinitely, so $\zeta_k = 0$.

Bound propagation and interpretation. The prescribed fault history and centering bias give the unfitted bound

$$\epsilon_k^D = \left\| D \sum_{\ell} h_\ell (f_{k-\ell} - f_k) \right\| + \|Db\|.\tag{111}$$

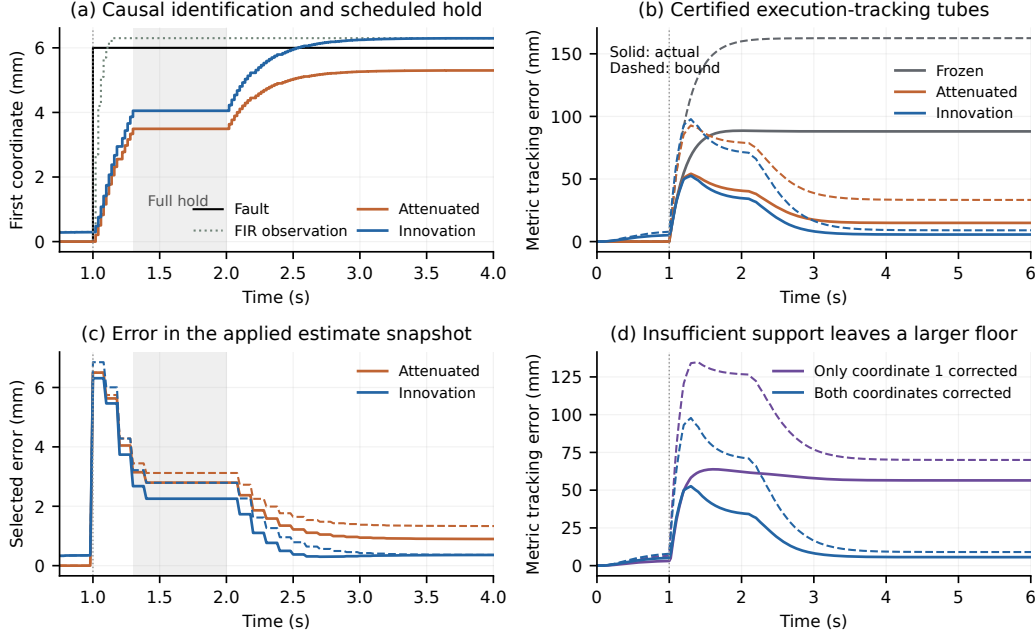


Figure 4: Constructed execution servo with a globally verified constant metric. Solid curves are realized errors or estimates; dashed curves in (b)–(d) are analytically propagated bounds. Shading denotes full-update holds. The FIR delays fault observation, and chunk snapshots delay application. Persistent centering bias gives a nonzero innovation floor; attenuation and missing correction support produce larger floors. Every bound is checked over the full 12 s rollout; panels show the recovery interval.

Use the exact selected fault increments in Proposition 1. Starting from $B_0^D = R_0 = 0$, propagate the actual-family bound, the snapshot minimum in Eq. (11), and

$$R_{k+1} = .92R_k + 2(\|(I - D)f_k\| + B_k^{D,app} + \|\zeta_k\|). \quad (112)$$

These are valid model-based bounds for this prescribed construction; fault truth is used only to evaluate their inputs, not by the adaptation law. The global metric removes a domain-containment restriction for this servo.

For a constant post-onset observation, the full-correction estimate limits are $\hat{f}_\infty = s_\infty(f + b)$ for attenuation and $\hat{f}_\infty = f + b$ for innovation, where $s_\infty = .841807$. The clock retains infinitely many active updates when extended periodically, and the box and $\|S\| = 1$ give $\alpha_k \geq .08/[1 + (2\sqrt{2})^2] = .0088$ on active updates. Finite snapshot age does not change these limits. The exact state-error limit is $\|(I - A)^{-1}(f - D\hat{f}_\infty)\|_P$. Its values, followed by the propagated envelope limits, are 88.03/162.50 mm (frozen), 14.94/33.29 mm (attenuated), 5.64/9.01 mm (innovation), and 56.46/70.00 mm (restricted support). The same fixed centering bias also causes a small healthy innovation drift before fault onset, visible in Figure 4b. The construction therefore illustrates useful recovery bounds without equating contraction with unbiased identification or complete authority.

The released `audit/verify_contraction_tracking.py` reproduces the figure, full trajectories, and assertions for the metric identity, causal observation, selected estimate error, applied snapshot, and state envelope. All assertions pass at absolute tolerance 10^{-11} (meters for error inequalities); the metric identity’s numerical spectral residual is 4.1×10^{-18} . Numerical checks validate this implementation of the analytical certificate; they do not certify any VLA rollout or establish full adaptive-system incremental contraction.

G EXPERIMENTAL PROTOCOLS AND ADDITIONAL RESULTS

All task-success counts in this appendix refer to simulation. Primary comparisons pair (task, initial state), without pinned policy sampling; a fixed episode changes failure to success, and a broken episode changes success to failure. Healthy runs measure nominal task competence. They are distinguished from faulted-frozen controls and from healthy runs with adaptation enabled. Early diagnostic tables without individual paired outcomes are reported using marginal counts only.

G.1 CALIBRATION, IDENTIFICATION, AND EVALUATION PROTOCOLS

Interface	Healthy calibration	Fault identification	Evaluation correction
LIBERO Panda	Three spatial episodes, 285 steps, with $\pi_{0.5}$; current plus six delayed commands	Within the evaluation episode	Episode-local updates; mask selected for the fault family
ALOHA	Eight episodes; joint-position predictor	One preceding faulted episode for the held protocol	Estimate held during 40 evaluation episodes
GR1	Six episodes; arm-position predictor	Reach windows, steps 50–200, in three preceding faulted episodes	Median estimate held during 30 evaluation episodes per condition

LIBERO reuses the predictor and sensitivity across suites, OpenVLA-OFT, and GR00T N1.7. ALOHA and GR1 each receive robot-specific calibration. Their held protocols include the listed identification episodes in addition to evaluation. Continuous-adaptation runs are separate conditions. Thus policy transfer on one interface and adaptation on a new interface use different calibration budgets.

Implementation provenance. A registered provenance pass recovered, for all nine cohorts, the code path, gate convention, clipping order, K , law constants, and calibration identity, archived as an implementation map with per-run configuration records. Every published cohort subtracts the correction outside the network; the native output-bias edit exists only in oracle and diagnostic scripts. The gate is a leakage deadzone on the LIBERO and ALOHA cohorts and a full hold on GR1; it never fired in the four headline samples or ALOHA’s identification episodes, and did fire in the joint-level cells, healthy controls, and the deadzone-.03 ablation row. $K = 6$ in every cohort. Known residual caveats are retained: the shipped sensitivity replay overran its source episode by five commands (the held-out calibration does not share this), two of twenty spatial episodes share a (task, init) pair with the plant log, constants were tuned on the same initial-state band, and the early unpaired four-suite tables used per-suite calibration where the paired headline uses the spatial one. The held-out cells answer the overlap caveats; no historical row is retrospectively reassigned to a cleaner update law.

The LIBERO primary condition estimates six motion coordinates and applies correction to the three rotation coordinates. Translation and rotation masks in the fault-family sweep were selected using faulted separation diagnostics. Healthy data fit the predictor and local sensitivity; fault-family development observations informed mask selection. A preregistered healthy-only gate (three-sigma phantom bands from twenty healthy estimate-only episodes) reproduces the rotation mask without faulted data; no automatic unknown-support selection beyond that gate is claimed. A model fit on healthy behavior also requires checking its error when the policy visits new contact and state regimes. The healthy predictor budget is tabulated above. LIBERO sensitivity uses central probes of $\pm.02$; total probe/reset counts, the fault-development selection budget, and complete task/seed identities for the final evaluations are not documented. Thus the entire development procedure cannot be described as healthy-data-only.

Architecture illustration and online information flow. Figure 1 depicts only the evaluated prediction-only information path. At step k , the frozen VLA supplies a_k and the available snapshot $\hat{f}_{\tau_k}$ is masked by D before realization and clipping. The resulting u_k is recorded at the modeled interface; the physical fault then produces $u_k^{\text{eff}} = u_k + f_k$. The post-command response y_k and the issued-command history enter the calibrated FIR predictor. Their residual is transformed by S^{-1} and healthy centering, then the declared attenuation or innovation rule updates or holds the estimate. The

resulting $\hat{f}_{k+1}$ can only affect a later command through the snapshot rule, so the diagram contains no same-step algebraic loop.

The mask D selects corrected coordinates and remains distinct from the residual normalizer P_r . Predictor coefficients H_ℓ , sensitivity S , healthy center b_{cal} , limits, and D are fixed before deployment. The figure is a mechanism diagram rather than empirical evidence: it contains no task-success cohort, nominal replay, task reward, fault truth, VLA update, geodesic signal, or online metric computation.

G.2 HEALTHY CENTERING AND PHYSICAL ERROR MEASUREMENTS

The model uses $z_k = S^{-1}e_k^{\text{res}} - b_{\text{cal}}$, with b_{cal} in the same coordinates as the estimated fault. Historical descriptions also call the baseline a settled healthy estimate. These definitions need not agree. For a constant healthy inverse-sensitivity observation b_0 , constant attenuation s_0 , and inactive projection, the uncentered attenuated filter settles at $s_0 b_0$. Subtracting that quantity before the same attenuation leaves a target $s_0(1 - s_0)b_0$, which is generally nonzero. With varying gains, $\mathbb{E}[s_k z_k] \neq \mathbb{E}[z_k]$ in general. Recomputing the gain after centering changes this illustrative expression again. Exact averaging quantity, units, subtraction location, and whether the normalizer is centered must therefore be recovered before assigning healthy drift to a physical fault. The existing counts are retained under their documented protocols.

For a simulator diagnostic, injected truth can reveal $w_k = z_k - f_k$, $\|D(\hat{f}_k - f_k)\|$, the uncorrected component $\|(I - D)f_k\|$, and the applied snapshot age. Truth is for evaluation, not deployment feedback. Decoder realization error must be isolated using base and edited actions at an identical observation and decoder draw; comparing different rollouts confounds realization with state change. Log projection and command clipping separately. Phase-resolved residual error, persistence and incremental-motion prediction baselines, and time until a prespecified execution tolerance is sustained can connect recovery to the theoretical budget. Those raw records are absent from the supplied experiment summaries, so no new VLA onset plot, physical-time recovery curve, or empirical metric estimate is reconstructed. The constructed example in Appendix F supplies such curves for its explicitly specified servo only. For execution-error measurements, replay the deployment sequence a_k through the nominal physical robot-controller model with matched initial conditions and exogenous inputs. The comparison ξ_k^{nom} uses those recorded commands; a healthy VLA rerun usually produces a different sequence and estimates task competence instead. A model-based replay also needs a valid model-error allowance. Controller state, physical delays, and the execution metric must describe the same subsystem.

G.3 STATISTICAL INTERPRETATION AND REPEATED EVALUATIONS

For b fixed and c broken episodes, the two-sided exact McNemar test is

$$p_{\text{McN}} = \min \left\{ 1, 2^{1-(b+c)} \sum_{j=0}^{\min(b,c)} \binom{b+c}{j} \right\}, \quad p_{\text{McN}} = 1 \text{ if } b+c = 0. \quad (113)$$

The primary four-suite comparison has $b = 51, c = 1$, giving $p = 2.354 \times 10^{-14}$ and an episode-weighted gain of $50/120 = 41.7$ percentage points. The unweighted mean of the four suite improvements is 44.4 points. Pointwise 95% Wilson intervals are $[16.7, 31.7]\%$ for 28/120 and $[56.1, 72.9]\%$ for 78/120; these describe the two marginal rates, not their paired difference. The four primary suite comparisons remain significant under either Holm or Bonferroni correction within that family of four tests.

For the paired effect, define $d_i = Y_i^{\text{corrected}} - Y_i^{\text{frozen}} \in \{-1, 0, 1\}$ and $\hat{\Delta} = (b - c)/n$. Under independent, identically distributed episode pairs, an estimated standard error is

$$\widehat{\text{se}}(\hat{\Delta}) = \sqrt{\frac{b+c - n\hat{\Delta}^2}{n(n-1)}}. \quad (114)$$

This follows from $\sum_i d_i^2 = b+c$ and the unbiased sample variance of d_i . For $n = 120, b = 51, c = 1$, the standard error is .04672; the normal approximation $\hat{\Delta} \pm 1.96 \widehat{\text{se}}$ gives $[-.3251, .5082]$. This is an approximate episode-iid 95% interval, not an exact interval or a task-clustered confidence statement.

The discordant counts also imply 27 pairs successful in both arms and 41 unsuccessful in both. Individual episode identities are unnecessary for this count-based calculation but are necessary to estimate task clustering or construct a task-level bootstrap.

A task-clustered bootstrap ($B = 20,000$, resampling the ten tasks per suite) gives pooled 95% intervals of $[+36, +65]$ episodes under the shipped calibration and $[+20, +49]$ under the held-out one, both excluding zero. Per suite, cluster-level sign-flip p under the shipped calibration is .031, .016, .061, .007 (spatial, object, goal, long-horizon) and under the held-out one .009, .081, .191, .127: three of four suites and one of four, respectively, against four of four at episode level. The pooled effect is robust to task clustering; individual suites at $n = 20$ –40 are not. Other condition-level tests are exploratory and nominal. Repeated evaluation of one recorded $\pi_{0.5}$ condition gives 13/20, 13/20, and 15/20, and the friction rerun’s identical twenty scenarios scored 8, 4, and 6 corrected across three runs, illustrating the variation in these small samples. Nonsignificance against healthy performance does not establish equivalence or noninferiority. Matched healthy controls on the exact four primary samples now exist: frozen 116/120 and 118/120 across two reruns (spatial 19–20, object 20, goal 39–40, long-horizon 38), and with the law running 117/120 and 118/120 ($b=4, c=3$ and $b=2, c=2; p = 1.0$), so the corrected rates are 67% (shipped) and 58% (held-out) of health on the same scenarios. The separate `libero_90` competence check gives 6/20 healthy successes and is excluded from the recovery headline.

Two severity conditions were repeated at $n = 40$ using the reported initial-state set 45–48. These are separate evaluations:

Fault	Frozen	Corrected	Fixed	Broken	Exact p
Rotation 0.05	27/40	39/40	12	0	4.883×10^{-4}
Translation 0.10	30/40	38/40	9	1	0.0215

The rotation-0.05 condition is significant in this repetition, whereas the original $n = 20$ cell has $p = .0625$. The translation repetition includes one regression. Zero regressions in the original severity cells therefore describe the observed sample rather than a general guarantee of harmless adaptation.

G.4 CALIBRATION AND THE MEASURED RESIDUAL

The LIBERO predictor uses the current command and six delayed commands. Its reported calibration-fit scores are:

Predictor R^2	dx	dy	dz	dr_x	dr_y	dr_z
Static	.885	.961	.978	.416	.091	.077
FIR, chart subtraction	.975	.982	.982	.489	.109	.168
FIR, relative rotation	.975	.982	.982	.522	.336	.972

Relative orientation uses $\text{Log}(R_{t+1}R_t^T)^\vee$ in a consistent frame. A numerical check over 2,000 random increments of magnitude 0.05 rad reports maximum error 1.4×10^{-14} , versus 6.28 for subtraction of axis–angle coordinates. The latter can cross a chart discontinuity. These fit and representation checks motivate the residual construction; they are not error bounds on unseen contact modes. For absolute-position outputs on ALOHA and GR1, a persistence predictor and incremental-motion error are necessary comparisons: high position R^2 alone does not identify command-induced motion, physical faults, or a contraction rate.

Per-axis central probes of ± 0.02 in the experiment’s action coordinates give

$$S = \begin{bmatrix} .297 & .021 & -.112 & -.029 & .081 & -.006 \\ .008 & .272 & .023 & -.034 & .016 & .003 \\ .029 & .032 & .126 & -.037 & .012 & -.003 \\ -.003 & -.001 & -.001 & .253 & -.004 & .002 \\ .013 & .001 & -.001 & -.007 & .276 & .000 \\ .001 & .004 & -.003 & .011 & .005 & .244 \end{bmatrix}. \quad (115)$$

The reported condition number is 3.0, and about 26% of the sum of absolute entries is off the diagonal. Repeating probes at ± 0.05 changes diagonal entries by ratios (.99, .92, 1.14, .99, .91, .99) and gives

reported condition number 2.2. This supports an approximately stable single-axis sensitivity over the probed *magnitude* range. Configuration is different: re-probing at held-out initial state 25 lowers the translation diagonal by 19–21% on y, z (rotation within 8%), and at a third initial state the z entry more than doubles (.126 $\rightarrow$.262) while the condition number falls from 3.0 to 1.29 – two of three registered stability bands failed. A single-state probe can carry a factor-of-two error in one axis, and the shipped matrix’s small z entry is a property of its probe state, not of the plant.

Simultaneous offsets produce larger deviations from superposition. The ratio of the measured uniform-fault response to the sum of single-axis responses is 0.77–1.04 at magnitudes 0.01 and 0.02. At magnitude 0.05, translation ratios depend on probe direction: x gives 0.07 and 0.78, and z gives 0.50 and 0.27. Contact and command limits are possible contributors; these measurements do not separate their individual effects. Corresponding separation measurements are:

Injected fault / measured channel	Estimated separation	Fraction of true offset
Uniform six-axis 0.05 / x	−0.0130	Wrong sign
Single-axis 0.05 / x	+0.0244	49%
Single-axis 0.02 / x	+0.0158	79%
Single-axis 0.02 / r_x	+0.0201	101%

Translation is therefore identifiable in some probed regimes even though the uniform-fault evaluation uses rotation-only correction. Local sensitivity, model error, and the realized fault family jointly determine useful correction directions.

G.5 SEVERITY MAP AND STRUCTURED OFFSETS

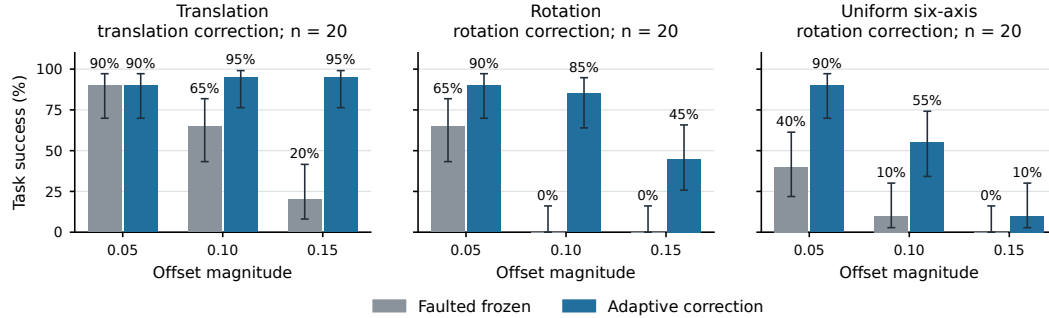


Figure 5: Reported severity sweep, LIBERO spatial: $n = 20$ paired episodes per condition. Bars show success; whiskers show marginal 95% Wilson intervals, not paired-effect intervals. Translation faults use translation correction; rotation and uniform faults use rotation correction. Supports are family-informed, not autonomously selected.

All cells in Figure 5 use 20 paired spatial-suite episodes. Translation faults use translation correction; rotation and uniform faults use rotation correction.

Family	Magnitude	Frozen	Corrected	Fixed	Broken	Exact p
Translation	.05	18/20	18/20	0	0	1
	.10	13/20	19/20	6	0	.03125
	.15	4/20	19/20	15	0	6.104×10^{-5}
Rotation	.05	13/20	18/20	5	0	.0625
	.10	0/20	17/20	17	0	1.526×10^{-5}
	.15	0/20	9/20	9	0	.003906
Uniform six-axis	.05	8/20	18/20	10	0	.001953
	.10	2/20	11/20	9	0	.003906
	.15	0/20	2/20	2	0	.5

The uniform-0.05 cell repeats the primary spatial condition. At uniform offset 0.15, the remaining translation disturbance and operation outside the calibrated regime are consistent with the limited

recovery. This comparison does not uniquely identify a failure mechanism. Figure 5 displays separate marginal Wilson intervals for each arm. For k successes among $n = 20$, $\hat{p} = k/n$, and $z = 1.959964$, the limits are

$$\frac{\hat{p} + z^2/(2n) \pm z\sqrt{\hat{p}(1-\hat{p})/n + z^2/(4n^2)}}{1 + z^2/n}. \quad (116)$$

These whiskers describe uncertainty in individual success rates under an episode-binomial model. They do not encode within-pair dependence or replace Eq. (114) for the primary paired difference.

For mixed-sign rotation offset $(+.06, -.06, +.03)$, success improves $2/12 \rightarrow 10/12$. The separation estimate is $(+.053, -.038, +.029)$: all signs agree with the fault, with magnitude errors 11.7%, 36.7%, and 3.3%. Zero-fault translation channels read $(-.003, -.024, -.006)$, showing nonuniform off-channel leakage. Discordant counts are unavailable for this exploratory condition, so no paired significance test is assigned.

G.6 ABLATIONS OF THE IMPLEMENTED OFFSET ESTIMATOR

Each row changes one setting relative to the LIBERO spatial reference: uniform $+0.05$ fault, rotation-only correction, $\gamma = 0.08$, $\rho = 0.15$, residual deadzone 0.008 , and a predictor fitted on three healthy episodes (285 steps). The reference repeats the primary spatial condition; its row is the archived headline file itself, not a fresh execution of the ablation runner. Endpoint estimates are last-50-step averages; healthy-subtracted separation estimates use a different measurement protocol.

Setting	Frozen	Corr.	Fixed	Broken	Exact p	Endpoint (r_x, r_y, r_z)
Reference	8/20	18/20	10	0	.00195	(.044,.020,.044)
$\gamma = .02$	8/20	17/20	9	0	.00391	(.036,.016,.036)
$\gamma = .32$	7/20	18/20	11	0	.00098	(.044,.019,.044)
$\rho = .05$	11/20	14/20	5	2	.453	(.027,.011,.027)
$\rho = .50$	8/20	20/20	12	0	.00049	(.048,.022,.049)
Deadzone 0	8/20	19/20	11	0	.00098	(.044,.018,.044)
Deadzone .03	8/20	17/20	9	0	.00391	(.031,.016,.032)
One healthy episode	10/20	17/20	8	1	.0391	(.027,.008,.044)
Two healthy episodes	8/20	17/20	10	1	.0117	(.045,.021,.043)

Success stays at 17/20 or higher over the tested $16 \times$ range of γ . Frozen totals differ across settings, so these within-row paired comparisons do not form a directly paired between-method ablation. Common task/initial-state/seed cohorts are needed to attribute differences to the changed setting alone. A smaller normalizer reduces estimated offsets and introduces two regressions in its sample. Under constant residual and inactive projection, the retention factor is $s = (1 + \|r\|^2/\rho^2)^{-1}$. At the representative residual norm 0.034 , it is 0.951 for $\rho = .15$ and 0.684 for $\rho = .05$. The latter differs from the approximately 55% observed retention on two axes. Residual variation, calibration bias, and finite-time behavior must also be accounted for to predict that magnitude. At $\gamma = .08$, 36 active updates reduce the transient toward a constant attenuated target by at least 95%. This does not mean a 95% reduction in error to the true fault. Attenuation could limit harmful correction when the predictor is inaccurate, but no matched attenuation-versus-innovation experiment establishes that mechanism.

G.7 TIME-VARYING OFFSETS

The following evaluations use amplitude 0.10 :

Profile	Frozen	Corrected	Reported exact p
Translation ramp over 60 steps	12/20	18/20	.031
Rotation oscillation, nonzero mean	13/20	20/20	.016
Intermittent rotation	12/20	19/20	.039

The tests for the ramp and intermittent profiles are retained as reported; their marginal counts do not determine the discordant pairs. Success on every corrected oscillatory episode determines seven fixed and zero broken pairs, giving $p = .015625$. Waveform RMS errors are 45–79% of fault amplitude. Useful correction therefore occurs despite imperfect waveform tracking in these conditions. A

zero-mean oscillatory condition leaves frozen performance essentially intact and provides no recovery evidence.

G.8 GAIN ESTIMATION AND THE INTEGRAL BASELINE

Replacing the instantaneous-command regressor by the FIR-weighted history reduces the healthy gain-estimation phantom from 0.181 to 0.020, and increases x -channel separation from 1.4 to 18.1 healthy-phantom standard deviations. The corresponding healthy success comparisons are $10/10 \rightarrow 9/10$ and $10/10 \rightarrow 10/10$. This diagnostic directly supports time alignment of the gain regressor.

Translation gain $g = .30$ gives $1/20 \rightarrow 17/20$ with reported $p = 3.1 \times 10^{-5}$. The estimates $(-.723, -.699, -.717)$ have absolute errors $(.023, .001, .017)$ relative to $\beta^* = -.700$, or $(3.3, .1, 2.4)\%$ of $|\beta^*|$. At $g = .20$, success improves $0/20 \rightarrow 17/20$, although some estimates saturate at the projection boundary. At $g = .50$, the comparison is $15/20 \rightarrow 18/20$ with reported $p = .25$. Rotation channels lacking the required excitation in the evaluated trajectories do not support gain-identification claims.

The raw-error integral baseline is $\hat{f} \leftarrow \hat{f} + k_i(y_t - a_t)$, evaluated under translation offset 0.15:

Integral gain	Frozen	Integral correction
.001	4/20	5/20
.005	3/20	7/20
.020	5/20	0/20
.050	2/20	1/20
.150	4/20	1/20
Model-based method	4/20	19/20

The best observed baseline success is 35%, compared with 95% for model-based correction, but frozen totals vary across this gain sweep. A seed-pinned paired rerun on one scenario set settles it: method 18/20 against integral 3/20, 15 fixed and 0 broken, $p = 6.1 \times 10^{-5}$. The estimates explain the gap – method 112/56/125% of the fault on $x/y/z$ against integral 18/41/163%, the integral’s x never arriving and its z overshooting. The comparison supports calibrating the command response; the separate static-observer cell isolates FIR memory, and a calibrated disturbance observer remains uncomparied.

G.9 NATIVE-BIAS REACHABILITY DIAGNOSTIC

An early $\pi_{0.5}$ experiment applies a computed six-vector to `action_out_proj/bias` under uniform +0.05 offset, scaled by k . Each condition uses 15 episodes:

Intervention	Successes
Faulted, $k = -1$	0/15
Faulted, $k = 0$ (no edit)	7/15
Faulted, $k = .5$	13/15
Faulted, $k = 1$ (computed oracle)	15/15
Faulted, $k = 1.5$	3/15
Healthy, $k = 1$	1/15

This diagnostic establishes a useful correction at the native bias site in that setup and substantial sensitivity to its sign and scale. It uses a different intervention and sample from the primary rotation-only comparison. A registered measurement later characterized the edit at fixed observations: deterministic and linear there (relative linear mismatch .035, quadratic part at most 8%), but with decoder gain varying 27–41% across observations, so a single mean-Jacobian bias vector realizes an intended $\pm .05$ correction with median relative error 43% (95th percentile 1.9, maximum 4.5 times the intended magnitude). This table therefore neither establishes exact external-action cancellation nor supplies an oracle upper bound for every evaluation protocol; the same-interface known-fault oracle in the main text does. Individual paired outcomes are unavailable.

G.10 FAULTS BELOW THE OPERATIONAL-SPACE CONTROLLER

These interventions act on the simulated Panda elbow below the controller. Correction uses translation channels with the healthy phantom subtracted – the one cohort that subtracts a settled attenuated estimate, measured open-loop and applied closed-loop. The historical friction and lock cells ran with model mutation on a cached environment and without a guaranteed common initial scene; a registered rerun repeats them with enforced model restoration and scenario reset.

Intervention	Frozen	Corrected	Fixed	Broken	Exact p
Torque bias 5 N m (historical)	20/40	32/40	15	3	.00754
Friction +4 (historical)	20/20	20/20	0	0	1
Friction +10 (historical)	11/20	14/20	6	3	.508
Friction +20 (rerun, $n=40$)	0/40	13/40	13	0	$2.44 \cdot 10^{-4}$
Lock within $\pm .05$ rad (rerun)	0/20	0/20	0	0	1
Healthy elbow, law running (rerun)	19/20	20/20	1	0	1

The frozen damage reproduces under the corrected protocol (friction 1/20 and 0/40 against the historical 0/20; lock 0/20), so the harm was not a protocol artefact. The friction repair did not reproduce at $n=20$ (4/20 against a registered keep-threshold of five); a preregistered $n=40$ extension resolved it at 0/40 $\rightarrow$ 13/40 with no regressions, replacing the historical 40% with a measured 33% from a floor of zero. The lock cell reproduces exactly, estimate included (final x, z estimates $-.232, -.189$ against the recorded $-.23, -.16$): identified, corrected, not repaired. On a healthy elbow the law holds a translation phantom of up to .167 and costs nothing. The torque comparison, on the historical protocol, also includes three regressions. The lock yields a residual separation of 15–23 healthy-phantom standard deviations but no recovery: strong detection does not ensure sufficient correction authority through the tested action interface; other controllers or replanned motions remain outside this comparison.

G.11 ADDITIONAL BACKBONES ON THE LIBERO INTERFACE

All rows use the unchanged LIBERO calibration. Offset rows use the indicated fault family; gain .20 acts on translation.

Policy	Suite	Condition	Frozen	Corrected	Fixed	Broken
OpenVLA-OFT	Spatial	Healthy	20/20	19/20	0	1
	Spatial	Rotation .10	0/20	11/20	11	0
	Spatial	Translation .15	0/20	14/20	14	0
	Spatial	Gain .20	0/20	17/20	17	0
	Object	Rotation .10	0/20	10/20	10	0
	Object	Translation .15	0/20	15/20	15	0
GR00T N1.7	Object	Gain .20	0/20	19/20	19	0
	Spatial	Healthy	18/20	18/20	1	1
	Spatial	Rotation .10	0/20	14/20	14	0
	Spatial	Translation .15	2/20	15/20	13	0
	Spatial	Gain .20	0/20	19/20	19	0

For OFT object-suite rotation, translation, and gain conditions, exact paired tests give $p = .00195$, 6.104×10^{-5} , and 3.815×10^{-6} , respectively. The healthy rows quantify regressions that aggregate rates can obscure: N1.7 retains the same total while changing two individual outcomes. Successful calibration reuse shows transfer over the evaluated trajectories; model error can still depend on the policy through state, contact, and excitation distributions.

G.12 ALOHA: IDENTIFICATION AND HELD CORRECTION

ALOHA transfer-cube uses 14 absolute joint targets at 50 Hz and 300-step episodes. Calibration on eight healthy episodes gives joint-position fit scores $R^2 = .989$ –1.000 on the twelve arm joints (the uncorrected right gripper fits at .77); the arm-block sensitivity condition number is 1.17, and the previously quoted 7.3 belongs to the full matrix including that gripper’s .145 diagonal. At left-arm offset $+.02$ rad:

Condition	Correction variation	Mean residual	Successes
Faulted frozen	—	2.12 cm	0/20
Continuous adaptation (attenuation law)	.43 cm	.17 cm	0/20
Static .019 rad correction	0	.11 cm	5/20
Prior identification, then hold	0	.07 cm	15/40
Faulted frozen, matched $n = 40$	—	—	0/40
Healthy, matched $n = 40$	—	—	17/40
Healthy, adaptation active	—	—	14/40

The held comparison has 15 fixed and zero broken episodes, giving $p = 6.104 \times 10^{-5}$. Its one preceding identification episode produces estimates .019–.020 rad, or 95–101% of the injected offset. The healthy active-law comparison has reported $p = .58$; discordant counts are unavailable.

Separation at offsets .05 and .02 rad is 85–91% and 94–106% of the fault, respectively, with leakage at most .007 rad on other joints. The .05 rad offset displaces the gripper by 5.3 cm. These observations show useful identification, while the continuous-adaptation condition illustrates that small mean residual alone does not ensure task success. A registered matched-initialization comparison then separated the confound: from the same stored estimate, the innovation law updating throughout repairs 14/40 against the held 15/40 ($p = 1.0$) while moving .30 cm within episodes, and the attenuation law repairs 0/40 while moving .40 cm – its estimate is pulled to .016 rad by step twenty (the attenuated fixed point plus deadzone leakage once the fault is cancelled), a .42 cm standing error during the grasp. The failure under continued adaptation is the attenuation law’s bias, not updating as such. LIBERO’s .019 figure is the mean per-step change of a velocity-offset correction (about .019 cm per commanded unit step, a realized arm-motion change of .0044 cm), while ALOHA’s .43 cm is the median within-episode range (SD .13 cm) of a position offset: the two are not on one scale, and the margin comparison across interfaces is qualitative. High absolute-position R^2 can also arise from slowly varying joint positions, and the registered diagnostic confirms it: the FIR’s position R^2 is within .001 of a persistence predictor’s. On position *increments* the held-out FIR reaches .53 (ALOHA left arm) and .38 (GR1 plate task) against persistence near zero, and -1.77 on the GR1 tray task, where it is worse than persistence; the correction relies on the plant’s steady-state gain, within a few percent of one on every corrected joint except the tray-task wrist (.91, .94).

G.13 GR1: FAULT MAGNITUDE AND TASK DEPENDENCE

The Fourier GR1 upper-body simulator uses GR00T N1.5, distinct from the N1.7 policy on LIBERO. There are 29 absolute joint targets at 20 Hz. Six healthy episodes fit the plate-task arm predictor with $R^2 \geq .998$; the tray-task cells use a separate ten-episode plant fitting at .994–.9997. Measured arm sensitivity is approximately .991. The residual norm excludes the twelve unmodeled hand joints. Identification uses reach-window measurements, steps 50–200, across three preceding faulted episodes; the median estimate is held during evaluation.

Task	Offset	Healthy	Frozen	Corr.	Fixed	Broken	Exact p
Plate	.05	19/30	7/30	24/30	17	0	1.526×10^{-5}
Plate	.10	22/30	1/30	19/30	18	0	7.629×10^{-6}
Plate	.20	25/30	0/30	21/30	21	0	9.537×10^{-7}
Tray	.10	17/30	1/30	13/30	12	0	4.883×10^{-4}

Offsets are in radians and apply to all seven working-arm joints. The four rows contain 120 evaluation pairs with zero observed regressions; the plate-.10 and tray-.10 rows repeat Table 4. Held estimates span 90–108%, 93–104%, and 94–99% of the true plate offsets, and 73–126% on the tray task. Continuous adaptation with the attenuation law on plate-.10 gives 8/30, versus 19/30 when held; in the registered matched-initialization rerun on one episode set, the innovation law continued gives 17/30 against 22/30 held, and the attenuation law 1/30 – on this robot updating itself costs five episodes, and holding is what the small task margin requires. Direct discordant pairs for the historical continuous-versus-held and healthy-versus-corrected comparisons are unavailable, so their significance is not reconstructed.

G.14 IDENTIFICATION TRACES AND TIMING SCOPE

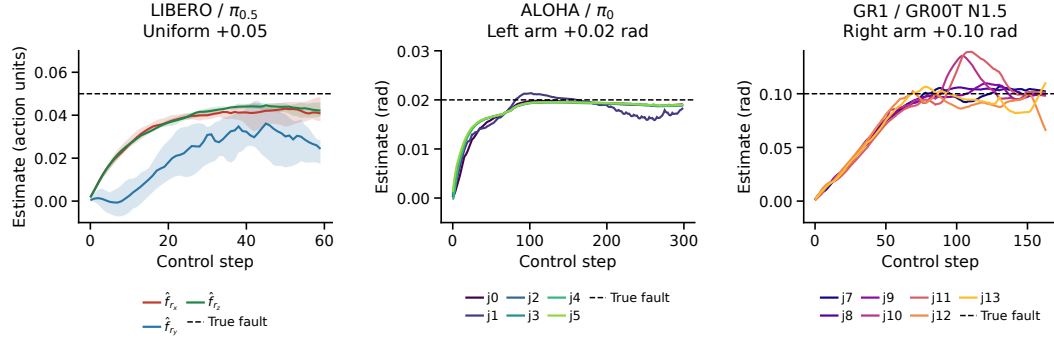


Figure 6: Reported identification traces in their original step coordinates. LIBERO shows the median and interquartile range over 20 evaluation episodes. ALOHA and GR1 show identification-episode traces, followed by separate held-correction evaluations. GR1 holds the median of reach-window estimates from three identification episodes. Dashed lines mark injected offsets; these plots do not report execution-recovery time.

Trace-level timestamps and exact fault-onset, gate, and correction-application events are not available. Accordingly, the horizontal axes are not converted into seconds, and inferred hold boundaries are not added. Nominal interface rates (50 Hz for ALOHA and 20 Hz for GR1) alone do not establish the update or policy-call rate represented by each logged step. Adapter latency is now measured over the eight healthy-control runs: .17 ms median, .46 ms at the 99th percentile, .77 ms maximum per control step, against a policy call near 900 ms and a simulator step of 30–40 ms on the evaluation GPU. Synchronized sensor-to-command delay, deadline misses, and a predefined sustained executed-action error threshold for recovery remain unmeasured; the traces show parameter identification behavior, not real-time guarantees.

G.15 MEASURED EXECUTION CERTIFICATES, ESTIMATOR RECURSIONS, AND THE r_y DEFICIT

The theory above is conditional on constants the paper originally left open. Each was preregistered with a refutation condition and then measured; the outcomes split cleanly between the estimator, where the recursions are predictive, and the plant, where the certificate the tube needs does not exist on the primary robot.

Same-command contraction. Eq. (13) was measured directly: a simulator state is saved mid-episode, perturbed, and the *identical* recorded command sequence is replayed from both states. Restoration is exact to 10^{-14} once the snapshot includes the solver warm start, actuator state, and the gripper’s accumulated target. Over 480 perturbations (three magnitudes, six directions, four save points) on the LIBERO Panda under OSC_POSE, the per-step gap ratio is 1.00 in every preregistered metric – joint, joint-and-velocity, end-effector, and end-effector in the tracking metric of the paragraph below – with 47% of free-space steps and 54% of contact steps at or above one. The controller sets each goal relative to the current pose, so a joint-space perturbation neither decays nor grows. Per the registered rule no further metric was tried. Contact raises joint-space expansion and damps end-effector gaps (32% of steps above one, against 50% in free space). Measured input sensitivity is .5–.6 cm per unit command on x, y, z , a factor 2.3–2.9 below the FIR’s first tap.

Servo certificates. Per-joint first-order fits (leave-one-episode-out, held-out increment R^2) certify the GR1’s seven right-arm joints and waist yaw, poles .72–.77 at 20 Hz with held-out R^2 .74–.95; the idle left arm and the hands fall outside. ALOHA at 50 Hz refutes the route on most joints – only two arm joints and the continuous gripper certify – because its 20 ms increments are noise-sized against the command–position gap. The Panda certifies per-axis *tracking* ($\lambda = .71, .67, .82, .83, .93, .61$ on x, y, z, r_x, r_y, r_z ; held-out R^2 .84–.99, stable to $\pm .02$ across three initial states), which is a response model and not the same-command contraction the tube requires; supplying it as the metric above still gives 1.00.

Tube and small-gain, propagated with measured constants. With $\lambda \approx 1$, $L = .11$ and $\eta = .18$ from these measurements, the tube of Theorem 2 covers the measured execution error on 100% of steps of 20 headline episodes with no violation to classify – and is vacuous: it never forgets, growing by η per step to a median $98\times$ the largest error in its own episode. The small-gain condition of Proposition 7 fails for the same reason and not for the one it is about: fitted coupling constants are small ($\epsilon_0 = .0008$, $k_E = .012$, $k_X = .009$), but $a = 1 - \lambda = 9 \times 10^{-5}$ gives $ac = 7 \times 10^{-6}$ against $bk_X = 9.6 \times 10^{-4}$. On this plant the composed certificate is unavailable, and the repair claim rests on the paired counts.

Estimator recursions. On the same telemetry the estimator-side predictions hold. The envelope of Proposition 1, propagated from the logged residual, gate and attenuation, covers the measured selected-coordinate error on 100% of steps. Fixed points predicted from the *logged residual distribution* $\mathbb{E}[s_k z_k]$ match the stored ablation settles within 20% on all nine (ρ , channel) pairs. Under the ramp profile the estimate’s lag (.016) sits inside $\nu/\underline{\alpha} = .021$. The registered onset statistic fails because the plant’s step noise is the size of the .01 step it must resolve, but the episode-mean response follows the FIR partial sums on r_x and r_z (plateau ratios 1.01 and .92) and overshoots by $3.0\times$ on r_y . The law’s own analysis is therefore predictive where the plant’s is not, and the attenuation/innovation distinction appears exactly as derived:

Settled estimate, fraction of the $+.05$ truth	r_x	r_y	r_z
Attenuation law (deployed)	.87	.41	.88
Innovation law, same scenarios	.95	.45	.98

Removing the attenuation removes the r_x, r_z shortfall, as Proposition 5 predicts, and leaves r_y where it was.

The r_y deficit is a calibration mismatch, not excitation or model order. Excitation and sensitivity are ruled out (M ’s r_y column is .25–.28 at three initial states, and r_y is commanded as much as r_z). A registered ARX(1) intervention, fitted on the same healthy log, repairs the headline spatial cell $9/20 \rightarrow 20/20$ but settles r_y at 32%, below the FIR’s 41% and the registered 60% floor: *refuted*. The cause is visible across plants. Every model fitted to the closed-loop healthy log recovers nearly the same r_y DC gain – .103 (FIR $K=6$), .118 ($K=20$), .099 (ARX) – against the open-loop probe’s .276, and the onset residual plateaus at .30 per unit offset, confirming the probe. The correction is executed at the probed gain but predicted at the fitted one, so the applied estimate feeds back into the residual and the fixed point sits low by that ratio:

	r_x	r_y	r_z
Predicted ratio (fitted DC gain)/(probed M)	.90	.36–.43	1.02
Measured settled fraction, across laws	.87–.95	.32–.45	.88–.98

A $2.7\times$ mismatch between closed-loop identification and the open-loop probe, on the channel with the slowest tracking pole (.93), therefore accounts for the shortfall on all three rotation channels. Constraining the fitted DC gain to the probed M , or identifying from open-loop segments, is the implied intervention; it is not registered and was not run.

Seed-pinned pairing. Pinning one sampler key per policy call makes two runs’ frozen arms agree on every outcome (80/80 and 20/20 pairs) and execute identical commands to the last step in 61/80 and 17/20 episodes, the remainder diverging at a policy-call boundary from GPU nondeterminism; two adaptive arms under different observers diverge at the first update, step 10–11. Paired comparisons within a pinned design therefore carry no sampling noise from the policy’s draws. Absolute levels do not transfer across designs: one key per call is a single fixed noise realization rather than the policy’s average, and the pinned method scores 5/40 on the long-horizon suite where the unpinned cell scores 15/40. Only within-design paired contrasts are compared here.

G.16 HARDWARE DIAGNOSTICS AND SENSING LIMITATIONS

The X2 humanoid record contains 134 physical rollouts with stiffness, damping, and feedforward scales from 0.1 to 1.3. The intervention is poorly visible in joint position and better reflected in effort,

2538 with reported correlation $r = .84$. The offset-detection range is $0.5\text{--}1.3^\circ$, and load-bearing joint
2539 behavior varies across runs. These are sensing diagnostics; no paired physical-robot repair result is
2540 included.

2541 Motion residuals can confound actuator and sensor biases. The checkpoint-specific zero-state-input
2542 ablation retains visual feedback and does not establish perceptual robustness. These diagnostic
2543 limitations delimit the tested physical interface.
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